" Combination of an Anti-PD1 antibody with Tisagenlecleucel Reinfusion in children, adolescents and young adults with Acute Lymphoblastic Leukemia after loss of persistence "

CAPTiRALL

CLINICAL TRIAL ON A MEDICINAL PRODUCT FOR HUMAN USE

Version N°4.0 of November 7th 2023
Project code number: APHP200132/EUDRACTN°: 2021-003035-28/ EUCT N°**2023-510121-15-00**

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INTERVENTIONAL RESEARCH PROTOCOL

PROTOCOL SIGNATURE PAGE

Research code number: APHP200132

Title: Combination of an Anti-PD1 antibody with Tisagenlecleucel Reinfusion in children, adolescents and young adults with Acute Lymphoblastic Leukemia after loss of persistence

Version N°4.0 of November 7th 2023

The trial will be carried out in accordance with the protocol, with current good practices and with statutory and regulatory requirements.

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List of abbreviations

ALL	Acute lymphoblastic leukemia
AYA	Adolescents and young adults
BM	Bone marrow
CAR	Chimeric antigen receptor
CNS	Central nervous system
CR	Complete remission
CRS	Cytokine release syndrome
CSF	Cerebro-spinal fluid
EFS	Event free survival
EMA	European medicines agency
FDA	US food and drug administration
GVHd	Graft vs host disease
HSCT	Hematopoietic stem cell transplantation
ICANS	Immune effector cell-associated neurotoxicity syndrome
IRAE	Immune related adverse event
MRD	Minimal residual disease
OS	Overall survival

TABLE OF CONTENTS

1. SUMMARY 10

1.1. SYNOPSIS OF CAPTiRALL 10
1.2. SUMMARY OF OBSERVATIONAL COHORT 16

2. SCIENTIFIC JUSTIFICATION FOR THE TRIAL 19

2.1. HYPOTHESIS FOR THE STUDY 19
2.2. EXISTING KNOWLEDGE RELATING TO THE CONDITION UNDER INVESTIGATION 19
2.3. SUMMARY OF RELEVANT PRE-CLINICAL AND CLINICAL TRIALS 19
2.4. DESCRIPTION OF THE POPULATION OF TRIAL SUBJECTS AND JUSTIFICATION FOR THE CHOICE OF SUBJECTS 20
INCLUSION CRITERIA: 20
2.5. NAME AND DESCRIPTION OF THE INVESTIGATIONAL MEDICINAL PRODUCT(S) 21
2.6. DESCRIPTION AND JUSTIFICATION OF THE DOSAGE, ROUTE OF ADMINISTRATION, ADMINISTRATION SCHEDULE AND TREATMENT DURATION 21

3. OBJECTIVES 23

3.1. PRIMARY OBJECTIVES 23
3.2. SECONDARY OBJECTIVES 23
3.3. OBJECTIVE OF ANY FUTURE ANCILLARY STUDY 24

4. DESCRIPTION OF THE TRIAL 24

4.1. CONCISE DESCRIPTION OF THE PRIMARY AND SECONDARY ENDPOINTS 24
4.1.1. CO PRIMARY ENDPOINTS 24
4.1.2. SECONDARY ENDPOINTS 24
4.2. RESEARCH METHODOLOGY 25
4.2.1. DESIGN OF THE TRIAL 25
4.2.2. NUMBER OF PARTICIPATING SITES 27
4.2.3. IDENTIFICATION OF THE SUBJECTS 27

5. PROCEDURE FOR THE TRIAL 27

5.1. SCREENING AND INCLUSION VISIT (WEEK-12, WEEK-1) 27
5.2. PROCEDURES OF THE TRIAL FOR BOTH COHORTS 28
5.2.1. FINAL COHORT ASSESSMENT AND PRE-TREATMENT (BETWEEN WEEK-3 AND WEEK-1) 28
5.2.2. LYMPHODEPLETING CHEMOTHERAPY (BETWEEN WEEK-2 AND WEEK-1) 28
5.2.3. TISAGENLECLEUCEL INFUSION (=DAY 0) 29
5.2.4. NIVOLUMAB ADMINISTRATION 29
A. CRITERIA TO DELAY NIVOLUMAB DOSING 30
B. CRITERIA TO RESUME NIVOLUMAB DOSING 30
C. CRITERIA TO PERMANENTLY DISCONTINUE NIVOLUMAB 30
D. MANAGEMENT ALGORITHMS FOR IMMUNE RELATED ADVERSE EVENTS 31
E. TREATMENT OF NIVOLUMAB RELATED INFUSION REACTIONS 31
5.2.5. FOLLOW-UP VISITS 32
5.3. TERMINATION AND EXIT RULES 46
5.3.1. TERMINATION OF THE STUDY 46
THE STUDY ENDS WITH THE LAST SCHEDULED VISIT OF THE LAST PATIENT INCLUDED. 46
5.3.2. CRITERIA AND PROCEDURES FOR PREMATURELY TERMINATING THE STUDY TREATMENT 46
5.3.3. CRITERIA AND PROCEDURE FOR PREMATURE WITHDRAWALS AND EXITS FROM THE STUDY 48
5.3.4. MONITORING SUBJECTS AFTER THE PREMATURE TERMINATION OF TREATMENT 48
5.3.5. FULL OR PARTIAL CANCELLATION OF THE STUDY 48

6. ELIGIBILITY CRITERIA 49

6.1. OBSERVATIONAL COHORT 49
6.2. INCLUSION CRITERIA 49
6.3. EXCLUSION CRITERIA 49
6.4. RECRUITMENT METHODS 50

7. TREATMENT ADMINISTERED TO STUDY PARTICIPANTS 50

7.1. THE INVESTIGATIONAL MEDICINAL PRODUCTS 50
7.1.1. INVESTIGATIONAL MEDICINAL PRODUCT 1 50
7.1.2. INVESTIGATIONAL MEDICINAL PRODUCT 2 51
7.2. ADDITIONAL MEDICINAL PRODUCTS 51
7.2.1. ADDITIONAL MEDICINAL PRODUCT 1 51
7.2.2. ADDITIONAL MEDICINAL PRODUCT 2 51
7.2.3. ADDITIONAL MEDICINAL PRODUCT 3 51
7.2.4. ADDITIONAL MEDICINAL PRODUCT 4 51
7.2.5. ADDITIONAL MEDICINAL PRODUCT 5 51
7.2.6. ADDITIONAL MEDICINAL PRODUCT 6 51
7.2.7. ADDITIONAL MEDICINAL PRODUCT 7 52
7.2.8. ADDITIONAL MEDICINAL PRODUCT 8 52
7.3. TRACEABILITY INFORMATION FOR THE INVESTIGATIONAL MEDICINAL PRODUCT(S) 52
7.4. AUTHORIZED AND PROHIBITED TREATMENTS (MEDICINAL, NON-MEDICINAL, SURGICAL), INCLUDING EMERGENCY MEDICATIONS: 52
7.4.1. AUTHORIZED TREATMENT DURING STUDY 52
7.4.2. PROHIBITED TREATMENTS 52
7.5. METHODS FOR MONITORING COMPLIANCE WITH THE TREATMENT 53

8. EFFICACY ASSESSMENT 54

8.1. DESCRIPTION OF PARAMETERS FOR ASSESSING EFFICACY ENDPOINTS 54
8.1.1. CO-PRIMARY END-POINTS 54
8.1.2. SECONDARY END-POINTS 54
8.1.3. EXPLORATORY END-POINTS 55

9. STEERING COMMITTEE FOR THE TRIAL 55

10. SAFETY ASSESSMENT - RISKS AND RESTRICTIONS ADDED BY THE STUDY 55

10.1. RECORDING AND REPORTING ADVERSE EVENTS 55
10.1.1. DEFINITIONS 55
10.1.2. THE ROLE OF THE INVESTIGATOR 56
10.1.3. SERIOUS ADVERSE EVENTS THAT REQUIRE A NOTIFICATION WITHOUT DELAY BY THE INVESTIGATOR TO THE SPONSOR 56
10.1.4. SPECIFIC FEATURES OF THE PROTOCOL 57
10.1.4.1. OTHER EVENTS THAT REQUIRE THE INVESTIGATOR TO NOTIFY THE SPONSOR WITHOUT DELAY 57
10.1.4.2. SERIOUS ADVERSE EVENTS THAT DO NOT REQUIRE THE INVESTIGATOR TO NOTIFY THE SPONSOR WITHOUT DELAY (EXCEPTION TO SAE RULES) 57
10.1.4.3. ADVERSE EVENT SPECIALLY FOLLOWED BY THE PROMOTOR 58
10.1.4.4. ADVERSE EVENTS DURING THE TRIAL POSSIBLY RELATED WITH THE TREATMENTS PRESCRIBED AS PART OF THE PATIENT'S STANDARD CARE 58
10.1.4.5. PERIOD DURING WHICH THE INVESTIGATOR MUST SEND NOTIFICATION OF SAES TO THE SPONSOR WITHOUT DELAY 58
10.1.4.6. PROCEDURES AND DEADLINES FOR NOTIFYING THE SPONSOR 59
10.1.5. ROLE OF THE SPONSOR 59
10.1.6. ANALYSIS AND DECLARATION OF SERIOUS ADVERSE EVENTS 59
10.1.7. ANALYSIS AND DECLARATION OF ADVERSE EVENTS SPECIALLY FOLLOWED BY THE PROMOTOR 61
10.1.8. ANALYSIS AND DECLARATION OF OTHER SAFETY DATA 61
10.1.9. ANNUAL SAFETY REPORT 61

10.1.10. DATA SAFETY MONITORING BOARD (DSMB) 61

11. DATA MANAGEMENT 62

11.1. RIGHT TO ACCESS SOURCE DATA AND DOCUMENTS 62
11.1.1. ACCESS TO DATA 62
11.1.2. SOURCE DOCUMENTS 62
11.1.3. DATA CONFIDENTIALITY 62
11.2. DATA PROCESSING AND STORAGE OF DOCUMENTS AND DATA 62
11.2.1. IDENTIFICATION OF THE DATA PROCESSING MANAGER AND THE LOCATION(S) 62
11.2.2. DATA ENTRY 62
11.2.3. DATA PROCESSING (CNIL, THE FRENCH DATA PROTECTION AUTHORITY) IN FRANCE 62
11.2.4. ARCHIVING 62
11.2.5. OWNERSHIP OF THE DATA 63

12. STATISTICAL ASPECTS 63

12.1. PLANNED STATISTICAL METHODS, INCLUDING THE TIMETABLE FOR ANY PLANNED INTERIM ANALYSES 63
12.2. HYPOTHESES FOR CALCULATING THE REQUIRED NUMBER OF SUBJECTS, AND THE RESULT 66
12.3. REFERENCES 66
12.4. STATE WHETHER SUBJECTS WHO EXIT THE STUDY PREMATURELY WILL BE REPLACED AND IN WHAT PROPORTION. 66
12.5. STATISTICAL CRITERIA FOR TERMINATION OF THE STUDY. 66
12.6. METHOD FOR TAKING INTO ACCOUNT MISSING, UNUSED OR INVALID DATA 66
12.7. MANAGEMENT OF MODIFICATIONS MADE TO THE ANALYSIS PLAN FOR THE INITIAL STRATEGY. 66
12.8. SELECTION OF POPULATIONS 66

13. QUALITY CONTROL AND ASSURANCE 67

13.1. GENERAL ORGANISATION 67

13.2. STRATEGY FOR SITE OPENING 67

13.3. SCOPE OF SITE MONITORING 67

13.4. QUALITY CONTROL 67

13.5. CASE REPORT FORM 67

13.6. MANAGEMENT OF NON-COMPLIANCES 68

13.7. AUDITS/INSPECTIONS 68

13.8. PRINCIPAL INVESTIGATOR'S DECLARATION OF RESPONSIBILITY 68

14. ETHICAL AND LEGAL CONSIDERATIONS 68

14.1. METHODS FOR INFORMING AND OBTAINING CONSENT FROM THE RESEARCH PARTICIPANTS 68

14.2. LEGAL OBLIGATIONS 69

14.2.1. THE SPONSOR'S ROLE 69
14.2.2. REQUEST FOR APPROVAL FROM THE INSTITUTIONAL REVIEW BOARD 69
14.2.3. REQUEST FOR APPROVAL FROM THE ANSM 69
14.2.4. DECLARATION OF COMPLIANCE WITH THE MR 001 "REFERENCE METHOD" 69
THIS RESEARCH IS GOVERNED BY THE CNIL (FRENCH DATA PROTECTION AGENCY) "REFERENCE METHODOLOGY FOR PROCESSING PERSONAL DATA USED WITHIN THE SCOPE OF HEALTH RESEARCH" (AMENDED MR-001). 69

14.2.5. MODIFICATIONS TO THE TRIAL 69
14.2.6. FINAL STUDY REPORT 69

15. FUNDING AND INSURANCE 70

15.1. SOURCES OF FUNDING FOR THE TRIAL 70

15.2. INSURANCE 70

16. PUBLICATION 70

17. BIBLIOGRAPHY 71

18. LIST OF ADDENDA 68

18.1. LIST OF INVESTIGATORS 68

18.2. CRS GRADING AND TREATMENT RECOMMENDATIONS 70

18.2.1. CRS GRADING SCALE ACCORDING TO ASTCT CONSENSUS 70
18.2.2. CRS MANAGEMENT GUIDELINES ACCORDING TO ELIANA TRIAL 70
18.2.3. ASTCT ICANS CONSENSUS GRADING SCALE FOR ADULTS 71

18.3. ENCEPHALOPATHY ASSESSMENT FOR CHILDREN ≥ 12 YEARS AND ADULT 72

18.3.1. ASTCT ICANS CONSENSUS GRADING SCALE FOR CHILDREN 72
18.3.2. ENCEPHALOPATHY ASSESSMENT FOR CHILDREN < 12 YEARS 73

18.4. MANAGEMENT ALGORITHMS FOR IMMUNE RELATED ADVERSE EVENTS 74

18.5. TREATMENT OF NIVOLUMAB RELATED INFUSION REACTIONS 77

CHANGE HISTORY

N°	Date of authorization	Changes
1	EC: 11/04/2023 CA: 16/02/2023	- Clarification on limiting toxicity between first nivolumab injection and D28 - Detail on follow-up period - nivolumab end of treatment allowed in case of lack response to treatment at least 3 months after tisagenlecleucel injection. - Detail on biological follow-up and biobanking in ancillary studies - Hospitalisation duration modification before tisagenlecleucel injection - Addition of investigating sites
2	EC : 21/09/2023 CA : 04/09/2023	- compliance with RCT with a view to transition to CTIS - Clarification of inclusion/non-inclusion criteria - Clarification of the procedures for pre-inclusion and inclusion of patients in the the trial - Modification of patient follow-up - Clarification of the procedures for temporary or definitive discontinuation - Clarification of the window in which visits must take place. - Clarification on monitoring response to treatment - Addition of a recruiting centre: AP-HM - Addition of a non-recruiting centre

1. Summary

1.1. Synopsis of CAPTiRALL

Full title	**Combination of an Anti-PD1 antibody with Tisagenlecleucel Reinfusion in children, adolescents and young adults with Acute Lymphoblastic Leukemia after loss of persistence**
Acronym	CAPTiRALL
Coordinating Investigator	Pr André Baruchel. Pediatric Hematology and Immunology – Department, Robert Debré University Hospital, Paris, France
Sponsor	Assistance Publique-Hôpitaux de Paris
Scientific justification	Tisagenlecleucel (CTL019) is an anti-CD19 autologous Chimeric Antigen Receptor (CAR) T-cell therapy, which has shown dramatic early results in advanced ALLs. Early loss of B-cell aplasia (recovery of B-cells in marrow/ peripheral blood within 6 months after infusion), a marker of the loss or non-functionality of the CAR T-cells, is associated to a very high risk of relapse. A reinfusion of CTL019, even after Fludarabine-Cyclophosphamide reconditioning, frequently fails to induce further expansion as observed in UPENN studies and in the Robert Debré Hospital experience. Non-persistence of CAR T-cells may be due to immune-mediated rejection or environment-mediated suppression of their growth. Evidence for increased PD-1 expression in CAR T-cells between infusion and peak expansion has been demonstrated in clinical samples. Preclinical data and few clinical data support a role of PD-1–PD-L1 blockade in improving the effectiveness of CAR T-cell therapy.
Main objectives and primary endpoint	To determine the safety, efficacy and feasibility of Nivolumab (Opdivo®)- an anti-PD1 treatment- combined to tisagenlecleucel in a cohort of relapsed or refractory B-ALL patients, aged 1-25 years old, previously treated by tisagenlecleucel (Kymriah®), with a demonstrated early loss of B-cell aplasia (within 6 months), a surrogate marker of the loss of CAR T-cells or their non-functionality. More specifically, the main objectives are: • **In cohort 1** that includes patients with a MRD negative disease status combined to an early loss (within 6 months) of B-cell aplasia To determine the optimal starting time of Nivolumab (Opdivo®) in terms of safety and efficacy among 4 candidate time points (day 14, day 11, day 5, and day -1). • **In cohort 2** that includes relapsed patients with an early loss (within 6 months) of B-cell aplasia To estimate the feasibility in terms of safety and efficacy of a very early start of nivolumab (day-1), prior to the reinfusion of tisagenlecleucel These objectives will be evaluated on the two following

	co-primary endpoints: **1.1 Limiting-toxicities*** between infusion and D28** **1.2 Efficacy, which evaluation is delayed at M3 after tisagenlecleucel reinfusion and defined by MRD negative CR* AND B cell aplasia**** *(*) MRD negative CR: undetectable disease at a 10^{-4} sensitivity threshold* *(**) B cell aplasia is defined by blood B lymphocytes < 10 /mm3 (or/and < 3% of total lymphocytes)* *(***)* Limiting toxicities are defined at day 28 after tisagenlecleucel infusion, by the occurrence of either ○ Related to CAR-T cells infusion: • CRS or aGVH: limiting toxicity will be defined as toxicity ≥ 4 • ICANs: limiting toxicity will be defined as toxicity ≥ 3 (excepted grade 3 seizure, ie focal, generalized seizure that resolves rapidly) ○ Related to nivolumab: immune related myocarditis, pneumonitis, encephalitis: limiting toxicity will be defined as toxicity ≥ 4
Secondary objectives and endpoints	Secondary endpoints: To assess the effect of these combined treatments on • Incidence of B cell aplasia at 6 months • Increase of B cell aplasia duration compared to the previous one observed after the first infusion of tisagenlecleucel (up to 24 months). • Disease best response • Complete remission (at M1 M3 M6 M12 after reinfusion) • Minimal residual disease (at M1 M3 M6 M12 after reinfusion) • 1-year OS • 1-year EFS • Incidence of Grade 3 adverse up to 2 years events such as - nivolumab-related adverse events: myocarditis, pneumonitis, encephalitis - tisagenlecleucel reinfusion-related events, in particular CRS or ICANs or aGVH, prolonged cytopenias • Incidence of Grade 3, 4 or 5 nivolumab-related adverse events up to 2 years: gut, liver, endocrine, stomatitis, rash or hematologic toxicity • Incidence of GVHD up to one year Exploratory endpoints: Long term efficacy: 2-year OS and EFS To explore the PD1 PDL1 axis and its correlation with success or failure

Design of the trial	This is a phase I/II clinical trial conducted in children, adolescents, and young adults with ALL, stratified on the disease status (marrow MRD negative or CD19+ relapse of B-cell ALL). Patients included first receive a lympho-depleting chemotherapy by Fludarabine / Cyclophosphamide. tisagenlecleucel infusion should then be administered 2 to 14 days after completion of chemotherapy. Nivolumab (Opdivo®) will be given intravenously at 3mg/kg every 2 weeks. **1) In the main cohort (cohort 1), i.e., in patients with MRD negative disease status: Time to Event Continual Reassessment Method (TITE-CRM) design** a. Four decreasing starting times for beginning nivolumab (day 14, day 11, day 5 and day -1) will be available for testing b. Patients will be enrolled sequentially by cohorts of 3 with escalation between cohorts based only on the limiting toxicities*** between infusion and D28 **2) For relapsed patients (cohort 2)**: Pilot trial, with nivolumab starting at day -1. **3) In the main cohort (cohort 1), i.e., in patients with MRD negative disease status: Time to Event Continual Reassessment Method (TITE-CRM) design** a. Four decreasing starting times for beginning nivolumab (day 14, day 11, day 5 and day -1) will be available for testing b. Patients will be enrolled sequentially by cohorts of 3 with escalation between cohorts based only on the limiting toxicities*** between infusion and D28 **4) For relapsed patients (cohort 2)**: Pilot trial, with nivolumab starting at day -1. Nivolumab will be given until 12 months after tisagenlecleucel infusion in case of response. *(***) Limiting toxicities are defined by the occurrence of any non- haematological targeted adverse events:* ○ Related to CAR-T cells infusion: • CRS or aGVH: limiting toxicity will be defined as toxicity ≥ 4 • ICANs: limiting toxicity will be defined as toxicity ≥ 3 (excepted grade 3 seizure, ie focal, generalized seizure that resolves rapidly) Related to nivolumab: immune related myocarditis, pneumonitis, encephalitis: limiting toxicity will be defined as toxicity ≥ 4
Category	*Cat 1/2 (Phase I/II)*
Population of trial subjects	Relapsed or refractory B-ALL (any relapse after HSCT, 2nd relapse or later, refractory ALL), aged 1-25 years old, previously treated by

	tisagenlecleucel (Kymriah®), Cohort 1 presenting early loss of B cell aplasia (< 6 months after infusion) And being in MRD negative complete remission or Cohort 2: presenting loss of B cell aplasia and having detectable marrow and/or blood and/or CSF CD19+ ALL cells.
Inclusion criteria	• Patients aged from 1 to 25 years (pediatric and young adults) with a history of CD19+ relapsed or refractory B-ALL (any relapse after HSCT, 2nd relapse or later, refractory ALL). • patient must have a second tisagenlecleucel (Kymriah®) product available • Cohort 1: previously treated by tisagenlecleucel (Kymriah®), and who present an early loss of B-cell aplasia defined by blood B lymphocytes < 10 /mm3 and/ or < 3% of total lymphocytes (< 6 months after infusion) while still being in CR with undetectable MRD • Cohort 2: previously treated by tisagenlecleucel (Kymriah®), who present a loss of B-cell aplasia defined by blood B lymphocytes < 10 /mm3 and/ or < 3% of total lymphocytes and a CD19+ ALL detectable disease in the marrow and/or Blood and/or CSF • Life expectancy > 12 weeks. • Karnofsky (age > 16) Lansky (age < 16) > 70 at screening. • No organ dysfunction • Who have signed an informed consent Affiliation to social security or any health insurance (as a beneficiary or assignee)
Exclusion criteria	• Patient has received intervening systemic therapy* for leukemia after first tisagenlecleucel infusion. (chemotherapy, anti leukemic immunotherapy, ITK, allogeneic HSCT) *systemic therapies exclude intrathecal therapy • Patient has an active autoimmune disease requiring systemic treatment within the past 2 years. • Patient has known history of, or any evidence of active, non-infectious pneumonitis. • Patient has a history of non-infectious pneumonitis that required steroid or has current pneumonitis. • Had receive prior therapy with an anti-PD1, Anti- PDL1 or anti-PDL2 agent. • Patient has hypersensivity to pembrolizumab/ nivolumab or one of its excipients • Patient has received a live vaccine injection within 45 days of planned start of trial therapy. • Patients with concomitant genetic syndromes associated with bone marrow failure states: such as patients with Fanconi anemia, Kostmann syndrome, Shwachman syndrome or any other known bone marrow failure syndrome. Patients with Down syndrome will

	not be excluded. • Patients with Burkitt's lymphoma/leukemia • Prior malignancy, except carcinoma *in situ* of the skin or cervix treated with curative intent and with no evidence of active disease. • Prior treatment with any gene therapy product except first tisagenlecleucel (Kymriah®) injection. • Prior treatment with any anti-CD19/anti-CD3 therapy, or any other anti-CD19 therapy, except for patients pre-treated with blinatumomab and/or tisagenlecleucel (Kymriah®). • Prior anti-cancer monoclonal antibody within 4 weeks before starting the trial. • Prior chemotherapy, targeted small molecule therapy, or radiation therapy within 2 weeks prior to reinfusion of CAR T cells or who has not recovered (i.e., ≤ Grade 1 or at baseline) from adverse events due to a previously administered agent. • Active or latent hepatitis B or active hepatitis C (test within 8 weeks of Screening), or any uncontrolled infection at Screening. • Human immunodeficiency virus (HIV) positive test within 8 weeks of Screening. • Presence of grade 2 to 4 acute or extensive chronic GVHD. • Active CNS involvement by malignancy, defined as CNS-3 per NCCN guidelines at the time of reinfusion of CAR T cells. Note: Patients with history of CNS disease that has been effectively treated will be eligible. • Uncontrolled acute life threatening bacterial, viral or fungal infection at Screening. • Previous or concurrent malignancy with the following exceptions: o Adequately treated basal cell or squamous cell carcinoma o *In situ* carcinoma of the cervix or breast, treated curatively and without evidence ofreccurrence of at least 3 years prior to the trial o *A primary malingnacy completelyresected and in CR for* ≥ 5 years • Pregnant or lactating women (female trial participants of reproductive potential must have a negative serum or urine pregnancy test performed within 48 hours before infusion) Patient with hypersensivity to Fludarabine and/or cyclophosphamide and/or tisagenlecleucel and/or nivolumab or one of their excipients.

Investigational medicinal product(s)	• Tisagenlecleucel (Kymriah®): second infusion of the product after initial manufacturing, if available, provided by NOVARTIS • Anti PD1 antibody: nivolumab (Opdivo®) provided by Bristol-Meyers Squibb
Number of subjects included	21 in cohort 1, 5 in cohort 2
Number of sites	11
Duration of the trial	- Total duration: 48 months - Recruiting period: 24 months - Follow-up phase: 24 months
Number of enrolments expected per site and per month	1 patient/center/month
Statistical analysis	**Sample size justification** **In MRD negative patients**: Such a TITE-CRM design has been shown to accurately estimate the optimal schedule based on sample sizes of 15-25 patients. A sample of 21 patients will be enrolled. **In relapsed patients**: The Pilot Trial will include a minimum of 5 patients and a maximum of 5 supplementary patients after interim analysis performed after the first 5 **Both designs** will use a Bayesian framework to determine the probability of success, response and tolerance (primary outcome). Sequential analyses will be performed Secondary endpoints will be estimated by maximum likelyhood method with 95% confidence intervals
Sources of funding for the trial	French Ministry of Health
Trial will have a Data Monitoring Committee	yes

1.2. Summary of observational cohort

Full title	**Observational cohort of patients with early loss of B- Cell Aplasia after 1st infusion of tisagenlecleucel**
Coordinating investigator	Pr André Baruchel. Pediatric Hematology and Immunology – Department, Robert Debré University Hospital, Paris, France
Sponsor	Assistance Publique – Hôpitaux de Paris
Scientific justification	Tisagenlecleucel (CTL019) is an anti-CD19 autologous Chimeric Antigen Receptor (CAR) T-cell therapy which has shown dramatic early results in advanced ALLs. Early loss of B-cell aplasia (recovery of B-cells in marrow/ peripheral blood within 6 months after infusion), a marker of the loss or non-functionality of the CAR T-cells, is associated to a very high risk of relapse. A reinfusion of CTL019, even after Fludarabine-Cyclophosphamide reconditioning, frequently fails to induce further expansion as observed in UPENN studies and in the Robert Debré Hospital experience. Currently, there is no consensus for management of these patients who are at high risk of relapse. Different strategies are used depending on physician decision: clinical monitoring, allogeneic hematopoietic stem cell transplantation, 2nd infusion of tisagenlecleucel or other. Here we would like to register prospectively patients who lost B-Cell aplasia within 6 months after 1st infusion to evaluate the eligibility rate of the CAPTiRALL trial evaluating the combination of an anti PD1 treatment and a second tisagenlecleucel infusion, and to describe the follow-up of patients presenting an early loss of B-cell aplasia and receiving the best current medical care.
Main objective and primary endpoint	The main objective of this observational cohort is to determine the eligibility rate of nivolumab (Opdivo®)- an anti-PD1 treatment- combined to tisagenlecleucel in a cohort of relapsed or refractory B-ALL patients, aged 1- 25 years old, previously treated by tisagenlecleucel (Kymriah®), with a demonstrated early loss of B-cell aplasia* (within 6 months), a surrogate marker of the loss of CAR T-cells or their non-functionality (CAPTiRALL trial). **B cell aplasia is defined by blood B lymphocytes < 10 /mm3 (and/or < 3% of total lymphocytes)*
Secondary objectives and endpoints	To register minimal medical data for all patients with relapsed or refractory B-ALL, aged 1-25 years old, previously treated by tisagenlecleucel (Kymriah®), with a demonstrated early loss of B-cell aplasia* within 6 months, a surrogate marker of the loss of CAR T-cells or their non-functionality. **B cell aplasia is defined by blood B lymphocytes < 10 /mm3 (and/or < 3% of total lymphocytes)* More specifically we will register medical decision according to physician, (ie, clinical monitoring, allogeneic hematopoietic stem cell transplantation, 2nd infusion of tisagenlecleucel or other) for each included patient and patient's disease status at the end of follow up.

Population of trial participants	Relapsed or refractory B-ALL (any relapse after HSCT, 2nd relapse or later, refractory ALL) patients, aged 1-25 years old, previously treated by tisagenlecleucel (Kymriah®), presenting early loss of B cell aplasia (<6 months after infusion) whatever disease status (MRD negative complete remission or with detectable marrow and/or blood CD19+ ALL cells) and whatever availability of a second tisagenlecleucel product.
Inclusion criteria	• Patients aged from 1 to 25 years (pediatric and young adults) with a history of CD19+ relapsed or refractory B-ALL (any relapse after HSCT, 2nd relapse or later, refractory ALL). • previously treated by tisagenlecleucel (Kymriah®), and who present an early loss of B-cell aplasia defined by blood B lymphocytes < 10 /mm3 and/or < 3% of total lymphocytes (< 6 months after infusion) while still being in CR with undetectable MRD. • Life expectancy > 12 weeks. • Karnofsky (age > 16) Lansky (age < 16) > 70 at screening. • Who have been informed about the trial and provide a non-opposition Affiliation to social security or any health insurance (as a beneficiary or assignee)
Exclusion criteria	• Patient presenting a detectable B-ALL (positive MRD or relapse) • Patients with concomitant genetic syndromes associated with bone marrow failure states: such as patients with Fanconi anemia, Kostmann syndrome, Shwachman syndrome or any other known bone marrow failure syndrome. Patients with Down Syndrome will not be excluded. • Patients with Burkitt's lymphoma/leukemia • Prior malignancy, except carcinoma in situ of the skin or cervix treated with curative intent and with no evidence of active disease. • Prior treatment with any gene therapy product except first tisagenlecleucel (Kymriah®) injection. • Prior treatment with any anti-CD19/anti-CD3 therapy, or any other anti-CD19 therapy, except for patients pre-treated with blinatumomab and/or tisagenlecleucel (Kymriah®). • Prior anti-cancer monoclonal antibody within 4 weeks before starting the trial. Prior chemotherapy, targeted small molecule therapy, or radiation therapy within 2 weeks prior to reinfusion of CAR T cells or who has not recovered (i.e., ≤ Grade 1 or at baseline) from adverse events due to a previously administered agent. • Previous or concurrent malignancy with the following exceptions: o Adequately treated basal cell or squamous cell carcinoma o *In situ* carcinoma of the cervix or breast, treated curatively and without evidence of

	recurrence for at least 3 years prior to the trial. ○ A primary malignancy completely resected and in CR for ≥ 5 years. • Pregnant or breastfeeding women
Number of participants included	1 patient/center/month
Number of centres	13 (12 recruiting centers and 1 non-recruiting center)
Duration of the trial	Total duration: 48 months - Recruiting period: 24 months - Follow-up phase: 24 months
Statistical analysis	**Cohort description** Clinico-biological characteristics will be described as usual: mean and standard deviation for quantitative variables, number and percentage for qualitative variables **Management of patients** Physician decision for management of patients will be described as usual: number, percentage and 95%CI for qualitative variables
Funding sources	French Ministry of Health

2. SCIENTIFIC JUSTIFICATION FOR THE TRIAL

2.1. Hypothesis for the Trial

After infusion of anti-CD19 chimeric antigen receptor (CAR) T-cells, in treatment of acute lymphoblastic leukemia (ALL), there is the next 15 days a dramatic expansion of the CAR T-cells followed by a decrease but in most of the cases persistence of the CAR T-cells. The B cell aplasia, linked to the killing of normal CD19+ B- lymphocytes by the CART-cells is used as a marker of their persistence. Moreover, a parameter associated to prolonged duration of ALL remission is precisely this persistence (1–3). Indeed, as in the ELIANA phase 1 trial, 22% of patients presented an early B cells recovery (before 6 months) and 15% relapsed with a CD19+ disease (Maude SL, ASCO 2018).

Strategies to improve the persistence of CAR T-cells have been developed such as the use of 4- 1BB co-stimulation domain, but persistence is not uniformly seen. Another cause of non- persistence of CAR T-cells could be an immune-mediated rejection of these cells due to the murine component of the CAR. Some groups are thus experimenting the use of humanized or fully human CAR constructs (2,4). A third but probably more important hypothesis is that non-persistence of CAR T-cells may be due to immune-mediated rejection or environment-mediated suppression of their growth for example through the PD1-PDL1 axis, opening a potential way to counteract it (2,5).

We hypothesize that the combination of anti-PD1 injection after tisagenlecleucel may improve CTL019 CAR T-cells reappearance and persistence in children and young adults treated for advanced acute lymphoblastic leukemia and who presented an early loss of CAR T-cells (within 6 months) after the first infusion with the objective to reduce the high relapse risk associated to this situation.

2.2. Existing knowledge relating to the condition under investigation

Nowadays, overall cure rates of acute lymphoblastic leukemia (ALL) approach 90%. Relapse rate is about 15-20% (6). Management of these relapsing patients is based on intensive chemotherapy, followed by allogeneic stem cell transplantation for early relapses. Prognosis decline with subsequent relapses: 2 years EFS range from 41+/-6% after second salvage therapy to 13-/+7% following third salvage therapy (6,7).

Tisagenlecleucel is an anti-CD19 CAR T-cells therapy that has shown dramatic early results in the treatment of advanced relapsed ALL in children and young adults. Tisagenlecleucel (Kymriah®) has been the first CAR T-cell therapy recently approved by the FDA and EMA (8), on the basis of the ELIANA and ENSIGN studies (1).

An update of the ELIANA study has been presented at the 2018 ASH congress: this phase 2 study including 92 children and AYA of whom 79 could be infused with Tisagenlecleucel, has shown an EFS of 66% at 18 months (IC95 52% to 77%) (1,9).

2.3. Summary of relevant pre-clinical and clinical trials

Tisagenlecleucel is an anti-CD19 CAR T-cells therapy that has shown dramatic early results in the treatment of advanced relapsed ALL in children and young adults. Tisagenlecleucel (Kymriah®) has been the first CAR T-cell therapy recently approved by the FDA and EMA (8). The ELIANA study, including 92 children and AYA of whom 79 could be infused with tisagenlecleucel, showed indeed an EFS of 66% at 18 months (IC95 52% to 77%) (9).

Initial expansion of CTL019, occurring within the first 15 days after infusion, is generally followed by the persistence of these cells. B cell aplasia is a marker of this persistence, impacting on remission duration. Twelve out of 60 patients (20%) included in a UPENN phase I study, presented an early B cell recovery (within 6 months' post-infusion). Seven out of these 12 pts relapsed (58%) (CD19+: 6; CD19-: 1) at a median time of 9 months (Maude S et al, ASCO 2017).

A reinfusion after early loss fails to induce further expansion even after Flu-Cy reconditioning as

observed in UPenn studies (S Maude, personal communication) or in the Robert Debré hospital experience (n=2).

Some preclinical studies support a role of PD-1–PD-L1 blockade in improving the effectiveness of CAR T-cell therapy as shown by Moon in 2016. In this preclinical trial, the use of an anti-PD1 antibody allowed to restore anti-NY-ESO-1 human transduced T cells efficacy against lung tumor cells by reducing tumor growth (35% smaller than in the control group) (10). Moreover, Gargett et al. showed that after repeated antigen stimulation, anti-GD2 CAR T-cells undergo activation- induced cell death depending on the level of antigen expression. The CAR T-cell viability could completely be restored by anti-PD1 repeated injections (11).

Regarding this encouraging preclinical results, anti-PD1 antibodies were used in CAR T-cells clinical setting. In a series of UPenn patients receiving pembrolizumab or nivolumab after second infusion of CTL019, 3 out of the 6 patients treated for early B cell recovery presented a B-cell aplasia for 5 to 15 months. Two patients out of the 4 patients who received anti-PD1 for extramedullary unresponsive disease presented a complete response and 2 a partial response (12)(1).
Of note, a phase II multicenter trial (NCT04225676) is actually in progress and aims to evaluate the efficacy and safety of tisagenlecleucel reinfusion in pediatric and AYA patients with B-ALL experiencing early loss of B-cell aplasia (13). Future results of this study could allow to evaluate the impact of adding anti-PD1 antibody to tisagenlecleucel reinfusion in this population.

2.4. Description of the population of trial subjects and justification for the choice of subjects

Description of the population
All patients receiving a 1st infusion of tisagenlecleucel for a B-ALL in France are included in the DESCAR T registry. After the infusion, B cell aplasia (BCA), a marker for CAR T-cells persistence/function are monitored every month until 6 months and then every 3 months until 12 months after infusion. In case of early loss of B cell aplasia (< 6 months after 1st infusion), relapse risk is higher and requires medical intervention.

Two related cohorts will be generated after loss of BCA has been confirmed:

- **Observational cohort**:
All patients, who received a tisagenlecleucel infusion, and who presented an early loss of B-cell aplasia (<6 months after 1st infusion) (B cell aplasia is defined by blood B lymphocytes < 10 /mm^3 or < 3% of total lymphocytes) with or without 2nd tisagenlecleucel product available will be registered in an observational cohort, unless opposition from the parents/child.
Minimal medical data (i.e., baseline data, treatment received, last-disease status, last follow-up) will be collected anonymously. Non-opposition of patients and or their parents/guardians is warranted before registering.

For the patients who will meet the following eligibility criteria, inclusion in the CAPTIRALL cohort will be proposed:

- **CAPTiRALL cohort:**

Inclusion criteria:
- All cohorts
    - Patients aged from 1 to 25 years (pediatric and young adults).
    - Relapsed or refractory CD19+ B-ALL (any relapse after HSCT, 2nd relapse or later, refractory ALL).
    - Life expectancy > 12 weeks.
    - Karnofsky (age > 16) Lansky (age < 16) > 70 at screening.
    - No organ dysfunction
    - All patients must have a second tisagenlecleucel (Kymriah®) product available.

- Cohort 1:

- Previously treated by tisagenlecleucel (Kymriah®), and who present an early loss of B-cell aplasia (< 6 months after infusion) while still being in CR with undetectable MRD

- Cohort 2:
  - Previously treated by tisagenlecleucel (Kymriah®), who present a loss of B-cell aplasia and a CD19+ ALL detectable disease in the marrow and/or blood and/or CSF

Justification: Tisagenlecleucel is approved by FDA and EMA for refractory or 1st relapse after allogeneic hematopoietic stem cell transplantation or second relapse or more CD19+ B ALL in patients aged from 1 to 25 years since 2017 in US and 2018 in Europe.

- An early loss of B cell aplasia is associated with a high risk of relapse, for the most part CD19+ relapse.
- Molecular relapse or morphological relapse after CAR T-cells is of dismal prognosis allowing to test a new therapeutic intervention
- Antibodies anti PD1 are already used in pediatric and young adult patients for refractory/relapsed solid tumors and hematologic malignancies like Hodgkin lymphoma or anaplastic large cell lymphoma.
- Immune related adverse events secondary to nivolumab are already known in these populations, which allow us to evaluate this product in these subjects.

2.5. Name and description of the investigational medicinal product(s)

- Tisagenlecleucel: additional dose after initial manufacturing, if available, provided by NOVARTIS
- Anti PD1 antibody: nivolumab (Opdivo®) provided by Bristol Meyers Squibb

2.6. Description and justification of the dosage, route of administration, administration schedule and treatment duration

- Lymphodepletion chemotherapy by fludarabine (30mg/kg/d D1 to D4) and cyclophosphamide (500 mg/kg/d D1 and 2) as previously described in ELIANA study (1).
- Additional dose of tisagenlecleucel infused intravenously 2 to 14 days after completion of lymphodepletion chemotherapy. A premedication by paracetamol (15 mg/kg) and dexchlorpheniramine (weight-adapted dose) should be given previously.
- Nivolumab (Opdivo®) will be given intravenously- after a paracetamol/dexchlorpheniramine- based premedication- at 3mg/kg every 2 weeks (maximum dose 240mg). Patients will receive nivolumab (Opdivo®) injection following subsequent adaptative design:

    - For cohort 1: four groups of decreasing treatment onset will be defined
        - Subcohort 1.14: Nivolumab will be started at D+14 after tisagenlecleucel injection.
        - Subcohort 1.11: Nivolumab will be started at D+11 after tisagenlecleucel injection.
        - Subcohort 1.5: Nivolumab will be started at D+5 after tisagenlecleucel injection.
        - Subcohort 1.-1: Nivolumab will be started the day before (D-1) tisagenlecleucel injection.
    - For Cohort 2: Nivolumab will be started the day before (D-1) tisagenlecleucel injection.

- Nivolumab (Opdivo®) will be continued until a maximum of 12 months post-infusion in case of response. In case of failure objectivized by limiting toxicities (grade ≥4 CRS or ICANs or aGVH at D28, B cells reappearance at M3 after tisagenlecleucel reinfusion or later, disease progression or immune related adverse events at any time nivolumab will be discontinued.
- The primary focus is the population of patients still in bone marrow MRD negative CR at the time of loss of B-cell aplasia (Cohort 1)

Nevertheless, a small exploratory cohort (Cohort 2) will be constituted by patients with CD19+ relapse and B-cell recovery. The proposal here is to advance the infusion of nivolumab (Opdivo®) at Day -1 of the CTL019 infusion to maximize the potential re-expansion of the CAR T-cells. The benefit-risk ratio

is considered adequate in this population with a dismal prognosis.

2.7 Summary of the known and foreseeable benefits and risks for the Clinical Trial participants

CAR T-cell therapy is complex and expensive, being currently administered in children and AYA with very advanced disease. The early loss of persistence of these cells, being associated to a very high risk of relapse, leads to a more or less desperate situation. It is thus essential to try to obtain a re-expansion of these cells. A second infusion followed by the administration of an anti-PD1 is a potential solution to this problem and could thus provide hopefully a curative option for these patients.

Tisagenlecleucel infusion after lymphodepletion chemotherapy is associated with grade 3 or 4 adverse events in 88% out of the 75 patients treated in the phase 1 study previously described (1).

- Lymphodepletion chemotherapy, and tisagenlecleucel infusion is known to induce severe pancytopenia. In the phase 1 study 53% of patients presented prolonged neutropenia and 35% presented a febrile neutropenia.
- Tisagenlecleucel infusion is associated with cytokine release syndrome (CRS) (77% of patients treated in the phase 1 study) often leading to intensive care unit admission. In the phase 1 study 1 patient died from cerebral hemorrhage in the context of coagulopathy and CRS. Patients will be monitored. They will be transferred in ICU and will received Tocilizumab if indicated.
- Various neurologic events are also described after tisagenlecleucel treatment (encephalopathy, delirium, agitation, somnolence, seizure). Patients will be closely monitored with daily neurological evaluation by ICANs scales as recommended (14).
- In patients who previously received a hematopoietic stem cell transplant, autologous CAR T-cells infusion expose to a potential risk of Graft versus Host disease (GVHd). Nevertheless, GVHd incidence after tisagenlecleucel infusion is very rare. In the phase I/II ELIANA study no acute nor chronic GVHd was described in the 18 patients (out of 30) receiving donor derived anti CD19 CAR T-cells (15). In an adults' cohort of 86 patients who survived in CR 1 year after CD19 CAR-T cells infusion, Cordeiro and al found a GVHd incidence about 20% (3 out of 15 patients who previously received HSCT) (16). In a larger review, Anwer and al., collected data of 72 patients from 7 studies, who received donor derived anti CD19 CAR T-cells for ALL or chronic lymphoblastic leukemia, or non-Hodgkin lymphoma. Only four patients presented an acute GVHd (grade 2-3) and 1 a chronic GVHd (17).

Using nivolumab (Opdivo®) like other anti-PD1 antibody increase risk of autoimmune disease or other drug-related adverse events.

- As described in Cochrane review anti-PD1 treatment is associated with drug-related adverse event such as fatigue (23%), diarrhea (15%), infusion reaction (14%), rash (12%) and such as colitis:enteritis (1%), endocrine dysfunction (14%), hepatotoxicity (5%), pneumonitis (4%)(18).
- In patients treated by nivolumab for 2 years for hematologic malignancies, adverse events of any grades occurred in 65% of cases, and grade 3 or more adverse events occurred in 22¨% of cases (19).
    - Immune-mediated adverse event occurred in 28 patients, predominantly grade 1 or 2 events. 15 patients had to stop nivolumab treatment.
    - Other drug-related adverse events were pneumonitis (n=9, grade 3 n=3), stomatitis (n=2), rash (n=7) or neutropenia (n=3). One case of grade 5 pneumonitis was described.
- Regarding pediatric patients, much less experience exists but nivolumab seems to be feasible and better tolerated. Indeed, in a phase II study including 44 children and young adults (aged bellow 30 years) with primary refractory Hodgkin lymphoma, were treated by a salvage therapy combining nivolumab and Brentuximab-Vedotin. Only 27% out of overall patients, and 32% out of patients aged under 18, experienced a grade 3-4 adverse event, most commonly neutropenia. No life- threatening adverse event was observed, and treatment did not have to be discontinued. Grade 1-2 adverse events such as maculopapular rash, and hypersensitivity were observed (20). In the brain tumors setting, Gorsi and al described a retrospective series of 10 patients treated for refractory pediatric brain tumors who received nivolumab combined with chemotherapy (n=10) and radiation (n=9). They observed only grade 2 nivolumab-related toxicity without dose-limiting side effect (21). Moreover, Kline and al published a retrospective cohort of 31 patients treated by re-

irradiation for recurrent intrinsic diffuse pontine glioma, from which 8 patients received nivolumab concomitantly. In this short series, only one case of hyponatremia was described, and resolved after nivolumab transient hold (22).

- Concerning anti PD1 antibody a multicohort phase II study evaluated safety and efficacy of nivolumab in relapsed/refractory classic Hodgkin lymphoma after failure of autologous stem cell transplantation in 243 adult patients (23). nivolumab seemed to be associated with a favorable safety profile with only 17 patients (7%) who discontinued treatment because of drug related adverse events, most commonly pneumonitis (2%) or autoimmune hepatitis (1%). Serious drug related adverse events occurred in 12% of patients. Fourteen patients (6%) discontinued treatment because of IRAE. However, most of IRAE resolved.
- More precisely nivolumab administered early after CAR T-cells infusion (D3) were reported in a series of 11 patients with relapsed/refractory B cell non-Hodgkin lymphoma with a safe toxicity profile (24). The infusions were safe, no dose limiting toxicities occurred. Grade 1 or 2 CRS was observed in 3/11 and 6/11 respectively and only 1 patient experienced reversible neurotoxicity.

Patients will also be closely monitored. Detection of any immune related adverse event will lead to premature termination of the treatment.

3. OBJECTIVES

The proposed trial aims to determine the safety, efficacy and feasibility of anti-PD1 injection in combination with tisagenlecleucel in a larger cohort of children and young adults, which will be treated prospectively and homogeneously. This should be the first French phase 1-2 trial using this innovative therapeutic combination.
Moreover, satellite studies will explore the PD1-PDL1 axis for possible correlations with success or failure.

3.1. Primary objectives

- To improve CTL019 CAR T-cell (Kymriah®) reappearance and persistence in patients treated for advanced acute lymphoblastic leukemia and who presented an early loss of B-cell aplasia (within 6 months), a surrogate marker of the loss of CAR T-cells or their non-functionality, after the first infusion with the objective to reduce the high risk of relapse associated to this situation.

More specifically, the main objectives are:

- In cohort 1 that includes patients with a MRD negative disease status combined to an early loss (within 6 months) of B-cell aplasia, to assess the optimal starting time of nivolumab (Opdivo®) in terms of safety and efficacy among 4 candidate time points (day 14, day 11, day 5, and day -1).
- In cohort 2 that includes relapsed patients with an early loss (within 6 months) of B-cell aplasia, to assess the feasibility in terms of safety and efficacy of a very early start of nivolumab, prior to the reinfusion of tisagenlecleucel

3.2. Secondary objectives

- To improve Event-Free Survival (EFS) in patients receiving a second infusion of tisagenlecleucel (Kymriah®) for advanced ALL
- To improve Overall Survival (OS) in patients receiving a second infusion of tisagenlecleucel (Kymriah®) for advanced ALL
- To evaluate feasibility of anti-PD1 (nivolumab-Opdivo®) infusion combined to tisagenlecleucel (Kymriah®) injection.
- To evaluate tolerance of anti-PD1 (nivolumab-Opdivo®) and CTL019 CAR T-cells in combination:
    - Incidence and grading of Cytokine Release Syndrome / ICANs. ASTCT scale will be used to grade CRS and neurotoxicity related to Kymriah® infusion.
    - Immune related adverse events (colitis, diarrhea, enteritis, endocrine dysfunction, hepatotoxicity, pneumonitis). All but CRS and Neurotoxicity adverse events will be graded using CTCAE v5.

- Acute or chronic GvHD incidence. Month Sinai Acute GVHd International Consortium (MAGIC) criteria will be used to grade acute GVHd and NIH2014 criteria for chronic GVHd.
- To explore the PD1-PDL1 axis and its correlation with success or failure

3.3. Objective of any future ancillary trial

1° Centralized satellite studies will explore the PD1-PDL1 axis for possible correlations with success or failure.
PD1-PDL-1 axis inhibition will be monitored by flow cytometry regarding PD1, LAG3 and Tim-3 expression.
2° CAR T-cell reappearance will be tracked by 2 methods already used in Robert Debré and Saint-Louis Hospital

- Flow cytometry using recombinant CD19 protein labeled through a biotin- streptavidin system will allow relative and absolute CAR T-cells quantification.
- Quantitative PCR will provide a confirmatory and more sensitive monitoring method by focusing on a CAR-specific genetic sequence, from genomic DNA extracted from whole blood.

3° CAR T-cell gene expression profiling will be studied at infusion and during treatment
4° Cytokines secretion will be monitored during treatment
5° Tumoral microenvironment of bone marrow will be studied at different time point: screening and M1, M3 at cellular and molecular level

4. DESCRIPTION OF THE TRIAL

4.1. Concise description of the primary and secondary endpoints

4.1.1. Co primary endpoints

1. Limiting-toxicities within the first 28 days after tisagenlecleucel

Limiting toxicities are defined at day 28 after tisagenlecleucel infusion, by the occurrence of either

- Related to CAR-T cells infusion:
  - CRS or aGVH: limiting toxicity will be defined as toxicity ≥ 4
  - ICANs: limiting toxicity will be defined as toxicity ≥ 3 (excepted for grade 3 seizure, i.e. focal, generalized seizure that resolves rapidly)
- Related to nivolumab: immune related myocarditis, pneumonitis, encephalitis: limiting toxicity will be defined as toxicity ≥ 4

2. Efficacy, defined at M3 after Tisagenlecleucel reinfusion, by MRD negative CR and B cell aplasia:

- Cohort 1
  - Subcohort 1.14: after 6 injections of nivolumab
  - Subcohort 1.11: after 6 injections of nivolumab
  - Subcohort 1.5: after 7 injections of nivolumab
  - Subcohort 1.-1: after 7 injections of nivolumab
- Cohort 2: after 7 injections of nivolumab

MRD negative CR is defined by undetectable disease at a 10^{-4} sensitivity threshold
B cell aplasia is defined by blood B lymphocytes < 10 /mm3 and/or < 3% of total lymphocytes

4.1.2. Secondary endpoints

- Incidence of B Cell aplasia at 6 months after infusion of tisagenlecleucel
- Increase of B cell aplasia duration compared to the previous one observed after the first

infusion of tisagenlecleucel

- Disease best response among
    - CR is defined by less than 5% of blasts in BMA (morphological remission)
    - MRD negative status is defined by undetectable disease at a 10^{-4} sensitivity threshold
    - Disease progression is defined as follows:
        - For Cohort 1
        - Reappearance of MRD-confirmed at 2 BM samples at a 2 week-interval
        - Any type of morphological disease (blood, marrow, CNS)
        - Clinical sign of CNS involvement (i.e., facial palsy)

        - For Cohort 2
        - if MRD+ at inclusion: increase by at least one log of MRD-confirmed by 2 BM samples at a 2 week-interval
        - if morphological relapse at inclusion: M2 or M3 marrow at D28 or later
        - Any type of morphological disease (blood, marrow, CNS)
        - Clinical sign of CNS involvement (i.e., facial palsy)

- Complete remission (measured at M1 M3 M6 M12 after reinfusion)
- Minimal residual disease (measured at M1 M3 M6 M12 after reinfusion)
- 1-year OS
- 1-year EFS
- Incidence of Grade 3 adverse events such as
    - nivolumab-related adverse events: myocarditis, pneumonitis, encephalitis
    - tisagenlecleucel reinfusion-related events, in particular CRS or ICANs or aGVH, prolonged cytopenias
- Incidence of Grade 3, 4 or 5 nivolumab-related adverse events: gut, liver, endocrine, stomatitis, rash or hematologic toxicity
- Incidence of GVHD

4.1.3 Exploratory end-points

Long-term efficacy measured on

- 2-year OS
- 2-year EFS

To explore the PD1 PDL1 axis and its correlation with success or failure

4.2. Research methodology

4.2.1. Design of the trial

This is a phase I/II clinical trial conducted in children, adolescents, and young adults with ALL, stratified on the status of the disease (MRD negative or CD19+relapse B-cell ALL).

1) In cohort 1: Time to Event Continual Reassessment Method (TITE-CRM) design

This is an adaptive sequential method initially proposed for dose-finding studies.
It is close to the standard 3+3 design

- Here, several starting times (day 14, day 11, day 5 and day -1) of the nivolumab (Opdivo®), will be available for testing (instead of doses in 3+3)
- It uses a sequential enrollment of patients by cohorts treated at the same starting point (instead of the same dose level in 3+3)
- It is ethically based: enrollment begins at the less exposed scheme (for the patient) – here the more delayed starting time (instead of the lowest dose level in 3+3)
- It is adaptive: it uses the data from the previous cohorts to modify the scheme for further cohorts

However, there are main differences that outperforms the CRM over the standard 3+3 design:

- It aims at defining a scheme of treatment associated with a targeted probability of limiting toxicity (here 0.2) - whereas 3+3 only targets a 0.3 probability of limiting toxicity
- It can use cohorts of various sizes (not necessarily of 3 patients and possibly varying) at each tested level
- It uses an underlying model relating treatment scheme and probability of toxicity, that allows an estimation procedure – whereas 3+3 only provides an optimal treatment scheme without any estimate of probability on the sole basis of empiric observation
- It uses all information recorded in the trial – while 3+3 only uses the last 3 (or 6) observations
- It includes in some sense an extension cohort in its own design, given the fact that it focusses its observations on the targeted treatment schemes (here starting doses) – while 3+3 only proposes an escalation scheme among the tested dose levels.

Last, the TITE-CRM is a variant of the original CRM [OQuigley1990] that handles late-onset toxicities. Contrarily to the original CRM that requires to wait until each patient completes the evaluation window of 28 days before using their outcome to forecast the best dose, it allows a new patient to be enrolled while all endpoints of the previous ones have not all been recorded (given the delay of 28 days before assessment).

The TITE-CRM design will be applied as follows:

Patients will be enrolled sequentially, by sub-cohorts of 3.
The first sub-cohort will receive the first dose of nivolumab at day 14.

- At day 28 after tisagenlecleucel infusion, each patient is evaluated in terms of immune related toxicity:
    - Related to CAR-T cells infusion:
        - CRS or aGVH: limiting toxicity will be defined as toxicity ≥ 4
        - ICANs: limiting toxicity will be defined as toxicity ≥ 3 (excepted for grade 3 seizure, ie focal, generalized seizure that resolves rapidly)
        -
    - Related to nivolumab: immune related myocarditis, pneumonitis, encephalitis: limiting toxicity will be defined as toxicity ≥ 4

- Note that, for ethical and safety concerns, inclusion of the second patient of each cohort will be delayed until evaluation at day 14 after first injection of nivolumab of the first patient from the same cohort for cohorts 1.14, 1.11,1.5 and at day 21 for cohort 1.- 1.
- Once a sub-cohort has been completed, based on the observed primary endpoints (i.e. occurrence or not of limiting toxicities within the first 28 days post infusion of tisagenlecleucel) of the 3 patients from this sub-cohort, the estimation procedure is rerun, and recommended allocation (ie, starting time of nivolumab) for patients in the second sub-cohort will use observed responses (limiting toxicity or not) of each previously enrolled patient. Note that the TITE-CRM allows the inclusion of a new patient within the 28 days of the last enrollment, with estimation accounting for such a delay.
- Note that for ethical concerns, no skipping will be allowed so that no starting time could be recommended unless all previous levels have been tested. Thus, patients from the second sub-cohort either will receive the first dose of nivolumab at day 14 (no change), or at day 11 (otherwise).

The process is continued until the sample size of 21 patients has been reached, unless stopping rules indicating futility high incidence of limiting toxicity(ies) (see details below in Statistical methods).

2) In cohort 2: Pilot trial based on 5 patients with CD19+ relapse to assess the feasibility of first administration of nivolumab at day -1

In absence of CRS or ICANs or a GVH grade ≥4, the second infusion will be done at day 14. Inclusion of the following patient will be delayed until evaluation at day 21 after first injection of nivolumab of the previous patient. After primary analysis, and if a response in observed for at least one patient, 5 supplementary patients could be included in this cohort 2.
In case of grade 5 toxicity, DSMB have to be asked to decide to continue (first injection at day -1), to adapt (delay the first injection at D5 after tisagenlecleucel infusion) or to interrupt this cohort 2 for further enrollment.

4.2.2. Number of participating sites

French sites will enroll patients (see Appendix 18.1).

4.2.3. Identification of the subjects

The subjects participating in this clinical trial will be identified as follows:
Site number (3 digits) - Sequential selection number for the site (4 digits) - surname initial - first name initial
This reference number is unique and will be used for the entire duration of the trial.

5. PROCEDURE FOR THE TRIAL

Before any examination or intervention may be carried out for the trial, the investigator must obtain the free, informed and written consent of the subject participating in the trial, or of his/her legal representative where applicable.

5.1. Screening, enrollment and inclusion visit (Week-12, Week-1)

Patients who received a first tisagenlecleucel (Kymriah®) injection for advanced ALL (see inclusion criteria), who presented an early loss of CAR T-cells (before 6 months after treatment) and for whom an additional dose is available are eligible.

Whose consent must be obtained	**Who informs the individual and collects their consent**	**When is the individual informed**	**When is the individual's consent collected**
The patient if aged more than 18 years. Guardians or parents in case of patient aged below 18 years. If patient is between 7 and 18 years old, assent should be looked for and could be materialized by a signature	*The local investigator or co-investigator*	*Screening visit*	*Patient and /or guardians will be given sufficient time to consent, at least 48h after the informed consent distribution.*

The investigator must not start any trial related procedure before the Informed Consent Form is signed anddated by both the patient (or both parents/legal representative, if applicable) and the investigator. Upon
signature of the Informed Consent Form, the investigator will confirm the registration of the patient for trial
participation sending the inclusion Confirmation form to the Clinical Research Unit.
The investigator or designee must ensure that only patients who meet all of the eligibility criteria are offered
enrollment in the trial. All data for the Inclusion/Exclusion criteria must be verifiable in the patient's source document.

At least 48h after the informed consent distribution patient and/or guardians consent have to be

collected.

Screening visit for pre-inclusion (enrollment of patient) will be performed after that the information on the availability of the additional dose of tisagenlecleucel product has been checked.

It could be performed during a short hospitalization or in outpatient setting with clinical, biological and radiological assessment for:

- A first disease status evaluation (BM aspiration with MRD evaluation, CSF collection, cerebral MRI) for determining initial cohort assessment (cohort 1 in case of negative MRD, cohort 2 in case of positive MRD or relapse).
- Confirmation of loss of B-cell aplasia and evaluation of CAR T cells detection and PD1-PDL1 axis
- Baseline blood test screening for verifying the absence of exclusion criteria (organ dysfunction or active auto-immune disease).

After the screening visit the procedure (ie day 1 of lymphodepleting chemotherapy) should start within the next 3 months.

Inclusion visit and final cohort assessment will be processed during a hospitalization at least 12 days before scheduled 2nd infusion of tisagenlecleucel with clinical, biological and radiological evaluation.

- A second disease evaluation will be performed (BM aspiration for MRD in CMF and qPCR, CSF lumbar collection) to determine the final cohort assessment (cohort 1 in case of negative MRD, cohort 2 in case of positive MRD or relapse) . Of note, if the last disease evaluation was performed less than 7 days before the inclusion visit this second evaluation is not mandatory.
- Confirmation of loss of B-cell aplasia and evaluation of CAR T cells detection and PD1-PDL1 axis
- Baseline blood test screening for verifying the absence of exclusion criteria (organ dysfunction or active auto-immune disease).

5.2. Procedures of the trial for both cohorts

A hospitalization of about 5 to 6 weeks is required for final cohort assessment, lymphodepleting chemotherapy and 2nd infusion of tisagenlecleucel combined with first infusions of nivolumab (at least 21 days after infusion of tisagenlecleucel and 12 days before). After the 21 days, patients should stay not far than 2 hours drive from center where Kymriah® was administered.

5.2.1. Final cohort assessment and pre-treatment (between week-3 and week-1)

Included patient will be hospitalized at least 12 days before the scheduled 2nd tisagenlecleucel infusion. The second disease evaluation will be performed (BM aspiration with MRD in CMF and qPCR, CSF lumbar punction) to determine the final cohort assessment (cohort 1 in case of negative MRD, cohort 2 in case of positive MRD or relapse). (see section 5.1)
Radiological and biological examinations will be performed to exclude infectious active disease. Note that, in case of active infection, especially in case of Sars-Cov2 positive PCR, lymphodepleting chemotherapy and tisagenlecleucel infusion must be postponed. Patients should have a Sars-Cov2 negative PCR before beginning the lymphodepleting chemotherapy.

Note that, in childbearing women, an effective contraception must be administered from the inclusion in the trial and until 5 months after the last nivolumab injection and/or until 12 months after the end of cyclophosphamide. Moreover, serum pregnancy test will be performed and must be negative before starting lymphodepletive chemotherapy.

5.2.2. Lymphodepleting chemotherapy (week-1)

The purpose of this chemotherapy is to induce lymphopenia in order to facilitate engraftment and homeostatic expansion of CAR T- cells. The lymphodepleting regimen is:

- Fludarabine (30 mg/m2 intravenously [i.v.] daily for 4 doses) and cyclophosphamide (500 mg/m2 i.v. daily for 2 doses starting with the first dose of fludarabine)

If there was previous Grade 4 hemorrhagic cystitis with cyclophosphamide, or the patient demonstrated a chemo-refractory state to a cyclophosphamide-containing regimen administered shortly before lymphodepleting chemotherapy, then the following will be used:

- Cytarabine (500 mg/m2 i.v. daily for 2 days) and etoposide (150 mg/m2 i.v. daily x 3 days starting with the first dose of cytarabine

After completion of lymphodepleting chemotherapy tisagenlecleucel infusion should be performed after 48 hours wash out period and within the next 14 days.

5.2.3. Tisagenlecleucel infusion (=day 0)

Patients will receive a 2nd tisagenlecleucel infusion (dose according to marketing authorization: 0.2 to 5.0×10^6 CAR positive viable T cells / kg for patients' ≤ 50 kg body weight or 0.1 to 2.5×10^8 CAR-positive viable T cells for patients > 50 kg body weight) and will be monitored for at least 21 days after the infusion during hospitalization.
Before infusion, the site must confirm that two doses of tocilizumab are on site prior to tisagenlecleucel infusion and one dose of siltuximab must be accessible within 24 hours of infusion for administration in order to manage suspected toxicities.
Side effects from T cell infusions can include fever, chills and/or nausea. All patients should be pre-medicated with acetaminophen or paracetamol and diphenhydramine or an H1 antihistamine. These medications can be repeated every 6 hours as needed. Nonsteroidal anti-inflammatory medication may be prescribed if the patient continues to have fever not relieved with acetaminophen or paracetamol. Steroids should NOT be used for premedication. It is recommended that patients NOT receive systemic corticosteroids other than physiologic replacement of hydrocortisone at any time, except in the case of life-threatening emergency or neurologic adverse event (see ICANs management in addenda 18.6), since this may have an adverse effect on CAR T- cell expansion and function.

The tisagenlecleucel dose will be administered via a single intravenous infusion. Depending on the volume of the tisagenlecleucel product, it will be given either as an i.v. infusion through a latex free i.v. tubing WITHOUT a leukocyte filter (approximately 10 – 20 mL per minute adjusted as appropriate for smaller children and smaller volumes) or as an i.v. push via a syringe (for smaller volumes). It is recommended that the infusion/i.v. push be completed within 30 minutes of thawing the cryopreserved product in order to preserve maximum cell viability. Vital signs (temperature, respiration rate, pulse, pulse oximetry, and blood pressure) will be taken prior to, during and immediately after the infusion and then approximately every 15 minutes for one hour and repeated at 2 hours. If vital signs are unsatisfactory and unstable, continue to monitor the patient until vital sign stabilization. All used infusion supplies, including the infusion bag and tubing, must be disposed of according to local institutional standard operating procedures.

5.2.4. Nivolumab administration

First infusion of nivolumab will be performed in hospitalization. Following infusions (after the minimal 21 days after tisagenlecleucel infusion) could be administered in an outpatient setting. To facilitate the schedule of treatment a table (§5.2.5) recapitulate the dates of nivolumab administration according to cohort attribution. Patient will receive nivolumab for at least 3 months after 2nd tisagenlecleucel infusion to evaluate efficacy of the combination.

- Nivolumab will be administered at a dose of 3 mg/kg (maximum dose=240mg), as a 30-minute IV infusion (+/- 5 minutes), every 2 weeks. A corrected dose must be recalculated at every administration, according to weight. All doses should be rounded to the nearest milligram.
- Nivolumab will be given until a maximum of 12 months after tisagenlecleucel infusion in case of response.
- A standard premedication will be applied according to local guidelines (based on dexchlorpheniramine and acetaminophen). If an acute infusion reaction is noted, subjects should be managed according to section e bellow.
- In both cohorts, subjects may be dosed with nivolumab no less than 12 days between doses and no more than 3 days after the scheduled dosing date (so an interval ranging from D12 to D17 after previous infusion). A nivolumab dose given after the 3-day window is considered as a dose delay. Nivolumab may be delayed for up to a maximum of 6 weeks from the previous dose.
- Subjects will be monitored continuously for IRAEs while on trial. Treatment modifications (e.g., dose delay or discontinuation) will be either based on specific laboratory and/or tisagenlecleucel related

AE and/or Immune Related Adverse Event (IRAE) criteria.

- Every nivolumab discontinuation and IRAE management grade ≥ 2 should be discussed with the coordinating investigator.
- Note that, in childbearing women, an effective contraception must be administered from the the inclusion the trial and until 5 months after the last nivolumab injection and/or until 12 months after the end of cyclophosphamide. Moreover, serum pregnancy test will be performed before each injection of nivolumab and then monthly until 5 months after the end of nivolumab and/or until 12 months after the end of cyclophosphamide.

a. Criteria to delay nivolumab dosing

Nivolumab administration should be delayed for the following:

- Any CRS or neurological event grade ≥ 2
- Any Grade 2 non-skin, IRAE, with the following exceptions:
  - Grade 2 IRAE fatigue do not require a treatment delay
  - Laboratory abnormalities (excluding bilirubinemia) do not require a treatment delay
- Any Grade 3 skin, IRAE
- Any Grade 3 drug-related laboratory abnormality, with the following exceptions:
  - Lymphopenia, leukopenia, neutropenia, anemia, thrombocytopenia of any grade do not require a treatment delay except for grade 4 neutropenia ≥ 3 months after tisagenlecleucel infusion which require discussion with the coordinating investigator

- Any IRAE, laboratory abnormality, or intercurrent illness which, in the judgment of the investigator, warrants delaying the dose of trial medication.
- Any criteria requiring definitive discontinuation

b. Criteria to resume nivolumab dosing

- Subjects may resume treatment with nivolumab when the IRAE(s), drug related laboratory abnormality or CRS/neurological events resolve to Grade ≤ 1 or baseline value, with the following exceptions: Subjects may resume treatment in the presence of Grade 2 fatigue
- Subjects who have not experienced a Grade 3 drug-related skin AE may resume treatment in the presence of Grade 2 skin toxicity
- Drug-related endocrinopathies adequately controlled with only physiologic hormone replacement may resume treatment.

If the criterion to resume treatment is met, the subject should restart nivolumab infusion. Total duration of nivolumab will nevertheless be limited to a maximum of 12 months after tisagenlecleucel infusion.
If treatment is delayed > 6 weeks, the subject must be permanently discontinued from trial therapy, except as specified in Section c.

c. Criteria to permanently discontinue nivolumab

Nivolumab treatment should be permanently discontinued for the following:

- Any Grade 2 drug-related uveitis or eye pain or blurred vision that does not respond to topical therapy and does not improve to Grade 1 severity within the re-treatment period OR requires systemic treatment
- Grade 3 drug-related uveitis, pneumonitis, bronchospasm, neurological event, hypersensitivity reaction, or infusion reaction of any duration requires discontinuation
- Grade 3 drug-related laboratory abnormalities should NOT require permanent discontinuation EXCEPT for total bilirubin > 5 x ULN and/or AST or ALT > 10 x ULN or Concurrent AST or ALT > 3 x ULN and total bilirubin > 2 x ULN
- Any other Grade 3 non-skin, drug-related AE lasting > 7 days
- Any Grade 4 nivolumab-related AE or laboratory abnormality, except for the following events which do not require a definitive discontinuation:
  - Grade 4 lymphopenia, leukopenia, anemia, thrombocytopenia
  - Isolated Grade 4 electrolyte imbalances/abnormalities that are not associated with clinical sequelae and are corrected with supplementation/appropriate management within 72 hours of their onset

- Grade 4 drug-related endocrinopathy AEs such as adrenal insufficiency, ACTH (Adrenocorticotropic Hormone) deficiency, hyper- or hypothyroidisms, or glucose intolerance, which resolve or are adequately controlled with physiologic hormone replacement (steroids, thyroid hormones) or glucose controlling agents, respectively, may not require discontinuation after discussion with the investigator
- Grade 4 neutropenia persisting 3 months after tisagenlecleucel infusion require discussion with the coordinating investigator

- Any dosing delay lasting > 6 weeks from the previous dose with the following exceptions:
  - Dosing delays to allow for prolonged steroid tapers to manage drug-related adverse events are allowed. Prior to re-initiating treatment in a subject with a dosing delay lasting > 6 weeks, the coordinating investigator must be consulted.
  - Dosing delays > 6 weeks that occur for non-drug-related reasons may be allowed if approved by the coordinating investigator. Prior to re-initiating treatment in a subject with a dosing delay lasting > 6 weeks, the coordinating investigator must be consulted.

- Any adverse event, laboratory abnormality, or intercurrent illness which, in the judgment of the Investigator, presents a substantial clinical risk to the subject with continued nivolumab dosing.
- Initiation of antineoplastic therapy (ie, chemotherapy, immunotherapy, radiation therapy).
- Nivolumab should be permanently discontinued in case of disease progression***** and after discussion with the coordinating investigator.

*****Disease progression is defined as follows:

For Cohort 1:

- reappearance of MRD-confirmed at 2 BM samples with 2 week-interval
- any type of morphological disease (blood, marrow, CNS)
- clinical sign of CNS involvement (i.e. facial palsy)

For cohort 2:

- if MRD+ at inclusion: increase by at least one log of MRD-confirmed at 2 BM samples with 2 week-interval
- if morphological relapse at inclusion: M2 or M3 marrow at D28 or later
- any type of morphological disease (blood, marrow , CNS)
- clinical sign of CNS involvement (i.e. facial palsy)

- Nivolumab should be permanently discontinued in case of secondary loss of response and after discussion with the coordinating investigator. Secondary loss of response is defined by:
  - B cell aplasia obtained at M3 after CAR T cells reinfusion and
  - Loss of BCA after M3 confirmed 2 times at 15 days interval.

Before the confirmation of the loss of BCA nivolumab should be continued.

In case of temporary suspension of treatment: the investigator must document the reason for suspending and resuming the treatment in the subject's source file and the case report form (CRF)

In case of Premature termination of treatment, if the subject remains enrolled in the trial until the end of the subject's participation: the investigator must document the reason in the subject's source file and the case report form (CRF)

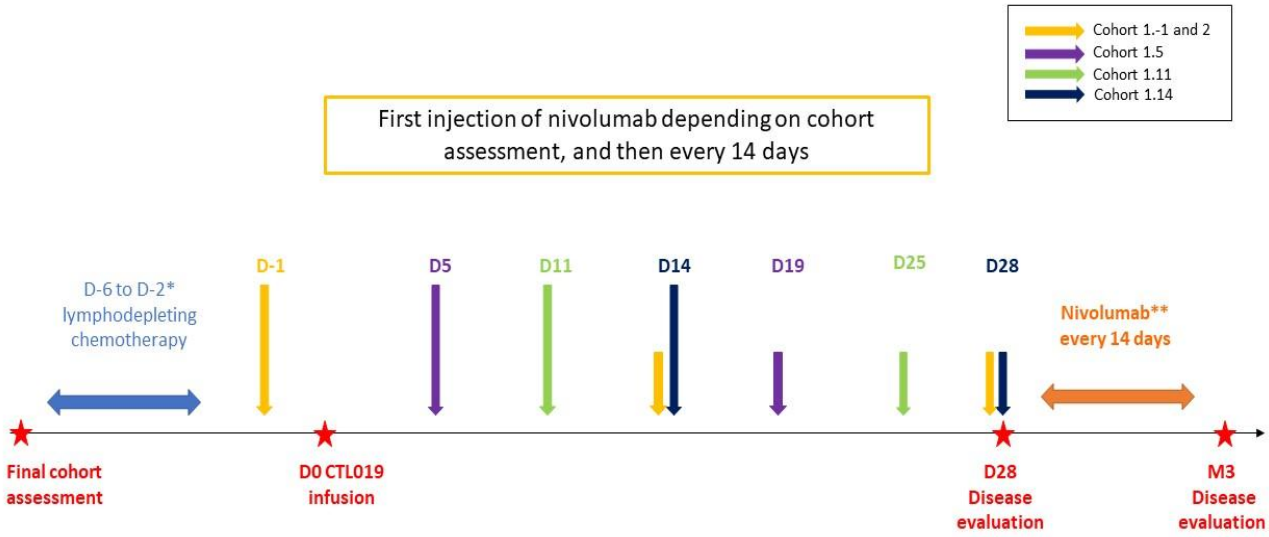
d. Management Algorithms for Immune Related Adverse Events

cf addenda 18.4

e. Treatment of Nivolumab Related Infusion Reactions

cf addenda 18.5

- **Time line of procedures of the trial from lymphodepleting chemotherapy to M3:**



CTL019 could be infused 2 to 14 days after completion of lymphodepleting chemotherapy
** Nivolumab could be administered +/- 3 days instead of the theoretical day

	Infusion of nivolumab#							Disease evaluation	Infusion of nivolumab
	1st	2nd	3rd	4th	5th	6th	7th		
subcohort 1.14	D14	D28	D42	D56	D70	D84		M3 +/- 3 days	everey 14 days until 12 months after Tisagenlecleucel infusion in case of success§
subcohort 1.11	D11	D25	D39	D53	D67	D81			
subcohort 1.5	D5	D19	D33	D47	D61	D75	D89		
subcohort 1.-1	D-1	**D14***	D28	D42	D56	D70	D84		
cohort 2	D-1	**D14***	D28	D42	D56	D70	D84		

* the second infusion in these cohorts will be delayed at D14
Criteria to delay and/or resume nivolumab infusion are described in §5.2.3
§ success is defined by MRD negative CR and B cell aplasia

5.2.5. Follow-up visits

After the initial hospitalization required for at least 21 days after the 2nd tisagenlecleucel infusion, outpatient evaluations will be performed every two weeks for all the follow up visits.
Patients will receive nivolumab infusion until at least 3 months after 2nd tisagenlecleucel infusion. Follow up visits are required to evaluate B-cell aplasia reappearance and duration, circulating CAR- T cells, occurrence of adverse events and to monitor disease response. Visits could be scheduled in a 2-days window around the theoretical date.

- Clinical and biological data should be collected at day of beginning lymphodepletion, D-1, D0, D3, D5, D7, D11, D14, D17, D21 and D28, and then twice in a month between D28 and 3 months after infusion and then every 3 months until 24 months after infusion (or 12 months after end of nivolumab).
- More specifically:
    - B cell aplasia will be closely monitored by flow cytometry with the following schedule: D-1, D0, D3, D5, D7, D11, D14, D21, and D28, and then twice in a month between D28 and M3, and then monthly until 6 months, and then every 3 months until 24 months after infusion.
    - Circulating CAR T-cells will be monitored by two methods (see table below) at D3, D5, D7, D11, D14, D17, D21, D28, and then twice in a month between D28 and 3 months after infusion. After 3 months, CAR-T cells will be monitored every month until 6 months, then every 3 months until 24 months.

- Occurrence of CRS or neurologic toxicity and their treatment should be registered. In case of CRS, following lab test will be performed: CRP, ferritin, triglyceride, chemistry, LDH, CBC, PT, aPTT, D dimer, fibrinogen, AST, ALT, bilirubin.
- Clinical evaluation of acute or chronic GVHd will be assessed using Month Sinai Acute GVHd International Consortium (MAGIC) criteria for acute GVHd and NIH2014 criteria for chronic GVHd as recommended by EBMT, NIH and CIBMTR consortium (25). Evaluations will be performed at D-1, D0, D3, D5, D7, D11, D14, D17, D21 and D28, and then twice in a month between D28 and D90 and then every 3 months until 24 months after infusion (or 12 months after end of nivolumab).
- Immune or non-immune adverse event related to nivolumab should also be closely monitored:
  - ACTH, cortisol at 8am, FSH, LH, oestradiol or testosterone, T4, proteinuria, antinuclear antibody, TPO ab, Tg ab: before 1st infusion and on demand according to clinical need
  - CBC, serum creatinine, sodium, potassium, calcium, magnesium, chloride, calcium, lipase, glucose, albuminemia, AST, ALT, total bilirubin, alkaline phosphatase before every infusion of nivolumab.
  - TSH at every month,

- **Disease response** will be monitored as follow:
  - For cohort 1 bone marrow aspiration (BMA) with cytologic and MRD assessment will be done at D28, M3, M6, M12 and M24 if no hematological abnormality.
  - For cohort 2 cytologic analysis and MRD on BMA will be assessed each month after infusion for the first 3 months and then at M6, M12 and M24 if no hematological abnormality.
  - Disease evaluation in CNS will be explored at D28, M3, M6, M12 and M24. In case of CNS involvement in screening evaluation, CSF analysis will be performed at D14 after CAR T cells reinfusion to assess disease response.

- Note that, all clinical evaluation should be realized in the investigation center. CAR-T cells related effect mainly occur during the first 28 days. It is also recommended that patients remain in the proximity of the investigation center (not more than 2 hours' drive) until D28 after Kymriah® infusion.

- In case of efficacy without dose-limiting toxicity, patients will be treated with nivolumab until 1 year after the 2nd tisagenlecleucel infusion and followed for a total of 2 years.

Expected length of participation, chronology and duration of the trial.

Length of Inclusion period	2 years
Duration of participation for each subject, of which:	
• Treatment period:	12 months.
• Follow-up period:	2 years after tisagenlecleucel infusion
Total trial duration:	4 years

To be noted: any patients who will received tisagenlecleucel during the CAPTIRALL clinical trial will have a 15 years follow-up in order to collect safety data. Such follow up will take place as part of an interventional clinical trial. Before the end of the CAPTIRALL clinical trial the sponsor will request an approval from the ethical committee and an authorization from competent authority.

Tables summarizing the chronology of the trial

Table 1: Clinical and biological explorations required from screening visit to M3 post-infusion

Actions	Screening visit (Baseline visit) Enrollment	Inclusion visit D-12 (final cohort assessment)	D-6 Lymphodepleting chemotherapy	D-1 Opdivo injection in cohort 2	D0 Tisagenlecleucel infusion	D3, D5, D7, D11, D14, D17, D21	every 15 days between D28 and M3
Information:	X						
Signature of the consent form	X						
Past medical history							
Demography	X						
Inclusion/exclusion criteria	X	X					
Medical history (including cardiovascular and pulmonary diseases)	X						
Diagnosis of cancer	X						
Cytogenetic of tumor / FISH/immunophenotyping	X						
Prior antineoplastic therapy	X						
Prior concomitant medication	X						
Clinical exam							
Physical exam	X			X	X	X	X
Height	X				X		X (each month)
Weight	X			X	X	X	X
Vitals signs	X			X	X	X	X
Tanner stage (< 18 years)	X				X		
Performance status assessment	X			X	X	X	X
Neurological status (expert examination)	X						
Hospitalization status	X			X	X	X	X

Actions	Screening visit (Baseline visit) Enrollment	Inclusion visit D-12 (final cohort assessment)	D-6 Lymphodepleting chemotherapy	D-1 Opdivo injection in cohort 2	D0 Tisagenlecleucel infusion	D3, D5, D7, D11, D14, D17, D21	every 15 days between D28 and M3
Imaging							
ECG	X				X		
Cardio ultra sonography	X			if clinically indicated			
Chest X Ray	X			if clinically indicated			
Infectious explorations							
Abdo-pelv ultra sonography			X	if clinically indicated			
Thoracic Scan			X	if clinically indicated			
Blood culture			X	if clinically indicated			
CBEU			X	if clinically indicated			
aspergillus antigenemia			X	X 1/week			
EBV, CMV, adenovirus PCR			X	X 1/week			
HIV test	X						
B and C Hepatitis tests	X						
A and B Influenza and SarsCoV2 PCR			X	if clinically indicated			
Tests (biochemistry, haematology)							
Haematology*	X		X	X	X	X	X
Chemistry**	X		X	X	X	X	X
Troponin	X			if clinically indicated			
During CRS only Ferritin, Triglycerid, Fibrinogen, PT, aPTT, D Dimer				X		X	X
IgG, A, M	X					X (D14)	X
Serum pregnancy test	X		X				X***

Bio Banking							
Serum			X		X	D7,D14, D21 and D1 of CRS	Only D28
Microenvironment (bone marrow)	X						Only D28 and M3
Lymphocyte sub-populations identification					X (before infusion)	D11	Only D28 and M3

*: Blood cells count (including red cells, white cells and platelets)
**: Blood Ionogram, liver enzyme, renal function
*** : pregnancy test before each nivolumab injection ie every 15 days until M12, must be negative prior to injection. Pregnancy tests should be performed monthly and until 5 month after the end of nivolumab and/or until 12 months after the end of cyclophosphamide

Actions	Screening visit (Baseline visit) Enrollment	Inclusion visit D-12 (final cohort assessment)	D-6 Lymphodepleting chemotherapy	D-1 Opdivo injection in cohort 2	D0 Tisagenlecleucel infusion	D3, D5, D7, D11, D14, D17, D21	every 15 days between D28 and M3
Disease assessment							
Bone Marrow aspiration	X	X#					X *
MRD in Bone Marrow by Flow cytometry	X	X#					X *
MRD in bone Marrow by qPCR	X	X#					X *
CD19 B cell population, and lymphocyte subpopulations by flow cytometry	X	X			X	X	X
CAR T-cells quantification (transgene)						X	x (one assessment in case of early loss of B cell aplasia)
Other tissues involvement assessment by biopsy	X						if clinically indicated *
CSF assessment by lumbar puncture	X	X#				X At D14 if CNS+at screening evaluation	at D28
CNS by MRI	X					if indicated	if clinically indicated
PD1-PDL1 axis assessment/extended phenotype of CAR (FCM)						X (D7, D14, D21)	X (only D28, M2 and M3)

* at D28 (and M2 for cohort 2) # not mandatory if the delay between the screening visit and the inclusion visit is shorter than 7 days

Actions	Screening visit	Inclusion visit D-12	D-6	D-1	D0 Tisagenlecleucel infusion	D3, D5, D7, D11, D14, D17, D21	every 15 days between D28 and M3
	(Baseline visit) Enrollment	(final cohort assessment)	Lymphodepleting chemotherapy	Opdivo injection in cohort 2			
Intervention							
Lymphodepleting chemotherapy			X				
Other chemotherapy while on trial	as clinically indicated						
Tisagenlecleucel infusion prerequisite assessment					X		
2nd CAR T cells (tisagenlecleucel) infusion					X		
Nivolumab first injection (and then every 15 days) **regarding cohort and sub group of patient**	**In the main cohort: Starting times for beginning nivolumab (Opdivo®) could be day 14, day 11, day 5 and day -1; and in the relapsed patients cohort starting time of nivolumab (Opdivo®) is D-1.**						
antineoplastic therapy after tisagenlecleucel infusion or trial discontinuation					X	X	X

Actions	Screening visit (Baseline visit) Enrollment	Inclusion visit D-12 (final cohort assessment)	D-6 Lymphodepleting chemotherapy	D-1 Opdivo injection in cohort 2	D0 Tisagenlecleucel infusion	D3, D5, D7, D11, D14, D17, D21	every 15 days between D28 and M3
Safety							
Adverse events CTCAE V5 grading			X	X	X	X	X
Anti-cytokine therapy assessment					X	X	
CRS and neurotoxicity grading					X	X	X
GVHD grading by MAGIC criteria (acute) or NIH2014 criteria (chronic)						X	X
Baseline endocrin status (ACTH, cortisol at 8am, FSH, LH, oestradiol or testosterone, T4)	X		X			if clinically indicated	
Baseline autoimmunity status (proteinuria, antinuclear antibody, Thyroperoxydase ab, Thyroglobulin ab)	X		X			if clinically indicated	
Anti-PD1 adverse event (lipase, glucose, albuminemia, AST, ALT, total bilirubin, alkaline phosphatase) **for patients who received at least one injection of Opdivo**	X					X	X
Monthly thyroid monitoring for patients **for patients who received at least one injection of Opdivo**	X						X

Table 2: Clinical and biological explorations required from M3 to M24

Actions	M3	M4, M5	M6	M9	M12	M15	M18	M21	M24
Clinical exam									
Physical exam	X	X	X	X	X	X	X	X	X
Height	X	X	X	X	X	X	X	X	X
Weight	X	X	X	X	X	X	X	X	X
Vitals signs	X	X	X	X	X	X	X	X	X
Tanner stage (< 18 years)	X	X	X	X	X	X	X	X	X
Performance status assessment	X	X	X	X	X	X	X	X	X
Hospitalization status	X	X	X	X	X	X	X	X	X
Imaging									
ECG									
Cardio ultra sonography					X				X
Chest X Ray									
Infectious explorations									
Abdo-pelv ultra sonography									
Thoracic Scan									
Blood culture									
CBEU									
aspergillus antigenemia									
EBV, CMV, adenovirus PCR	X	X	X	X	X				
A and B Influenza A and B tests, SarSCoV2 PCR									

Actions	M3	M4, M5	M6	M9	M12	M15	M18	M21	M24
Tests (biochemistry, haematology)									
Haematology*	X	X	X	X	X	X	X	X	X
Chemistery**	X	X	X	X	X	X	X	X	X
During CRS only Ferritin, Triglycerid, Fibrinogen, PT, aPTT D Dimeres									
IgG, A, M	X	X	X	X	X	X	X	X	X
Serum pregnancy test	Each 15 days between M3 and M12: a negative pregnancy test is required prior to each nivolumab injection ***								
HIV test									
B and C Hepatitis tests									
Disease assessment									
Bone Marrow aspiration	X		X	X	X				X
MRD in Bone Marrow by Flow cytometry	X		X	X	X				X
MRD in Bone Marrow by qPCR	X		X	X	X				X
CD19 B cell population, and lymphocyte subpopulations by flow cytometry	X	X	X	X	X	X	X	X	X
CAR T quantification (transgene)	X (one assessment in case of early loss of B cell aplasia)			X	X	X	X	X	X
Other tissues involvement assessment by biopsy	if clinically indicated								
CSF assessment by lumbar puncture	if clinically indicated								
CNS by imaging (cerebral MRI)	if clinically indicated								
PD1-PDL1 axis assessment/extended phenotype of CAR (FCM)	X*	X*	X*	X*	X*	X*	X*	X*	X*

*In case of persistence of CAR T cells detection

Actions	M3	M4, M5	M6	M9	M12	M15	M18	M21	M24
Intervention									
antineoplastic therapy after tisagenlecleucel infusion or trial discontinuation	X	X	X	X	X	X	X	X	X
Safety									
Adverse events CTCAE V5 grading	X	X	X	X	X	X	X	X	X
Anti-cytokine therapy assessment									
CRS and neurotoxicity grading									
GVHD grading by MAGIC criteria (acute) or NIH2014 criteria (chronic)	X	X	X	X	X	X	X	X	X
Baseline endocrin status (ACTH, cortisol at 8am, FSH, LH, oestradiol or testosterone, T4)	if clinically indicated								
Baseline autoimmunity status(proteinuria, antinuclear antibody, Thyroperoxydase ab, Thyroglobulin ab)	if clinically indicated								
Anti-PD1 adverse event (lipase, glucose, albuminemia, AST, ALT, total bilirubin, alkaline phosphatase **for patients who received at least one injection of Opdivo**	X	X	X	X	X	if clinically indicated			
Monthly thyroid monitoring (TSH **for patients who received at least one injection of Opdivo**	X (monthly)					if clinically indicated			

*: Blood cells count (including red cells, white cells and patelets)
**: Blood Ionogram, liver enzyme, renal function
*** After the last injection of nivolumab, pregnancy tests should be performed monthly and until 5 months after the end of nivolumab and/or until 12 months after the end of cyclophosphamide.
° CAR-T cells will be tested until they are not detectable

Table 3: Clinical data collected for the observational cohort

Actions	**Screening visit** (**Baseline visit)**	**Last Follow-up**
Information:	X	
Signature of the non-opposition form	X	
Past medical history		
Demography	X	
Inclusion/exclusion criteria in CAPTiRALL	X	
Non eligibility criteria for CAPTiRALL	X	
Medical history	X	
Diagnosis of cancer	X	
Cytogenetic of tumor / FISH/immunophenotyping	X	
Prior antineoplastic therapy	X	
Prior concomitant medication	X	
Medical data		
Chemotherapy received		X
Allogeneic stem cell transplant		X
Immunotherapy		X
Disease status		X
Live status		X

- **Distinction between standard care and research**

Table 4: "Standard care" vs. "added interventions" required specifically for the study

Procedures and treatments to be provided during the trial	**Procedures and treatments associated with <u>standard care</u>**	**Procedures and treatments added for the <u>trial</u>**
Treatment		Tisagenlecleucel combined with nivolumab
Consultations	Every month until 6 months after 1st infusion of tisagenlecleucel then every 3 months. Disease assessment: - bone marrow aspiration every month until 3 months after 1st infusion of tisagenlecleucel then every 3 months -lumbar punction 1 month after 1st infusion of tisagenlecleucel	D-1, D0, D5, D7, D11, D14, D17, D21 and D28, then twice a month between D28 until the end of nivolumab treatment and then every 3 months until 12 months after end of treatment by nivolumab.
Blood samples	Haematology, chemistry, CRS assessment, disease assessment, IgG,A,M, Flow cytometry using biotin-conjugated soluble CD19 protein will allow relative and absolute quantification of CAR T- cells, as well as their phenotype characterization. Quantitative PCR will provide a confirmatory and more sensitive monitoring method by using a CAR-specific genetic sequence, from genomic DNA extracted from whole blood.	Anti PD1 adverse event detection, PD1-PDL1 axis exploration Serum Pregnancy test at baseline visit, D-6 tisagenlecleucel infusion and every 15 days between D28 and M3.
Imaging		Thoracic imaging, ECG, cardio ultrasonography, abdomino-pelvien ultrasonography, cerebral MRI

Detection of circulating CAR-T cells by Flow cytometry or quantitative PCR are a part of standard care of patients who received a Tisagenlecleucel infusion. Volume of blood samples required by the trial are calculated in the tables below and are in compliance with European ethical considerations for clinical trial on medicinal products.

Poids < 30kg	J1 LD	J0	J3	J5	J7	J11	J14	J17	J21	J28	CRS	M1,5	M2	M2,5	M3
dosage cytokines	3	3			3		3			3	3				
phénotype éténdu					3		3			3			3		3
sous pop lymphocytaire et transcriptome		3				3				3					3
total	3	6	0	0	6	3	6	0	0	9	3	0	3	0	6
total par 4 semaines	5	**30**									3	3		6	

Poids >=30kg	J1 LD	J0	J3	J5	J7	J11	J14	J17	J21	J28	CRS	M1,5	M2	M2,5	M3
dosage cytokines	5	5			5		5			5	5				
phénotype éténdu					5		5			5			5		5
sous pop lymphocytaire et transcriptome		5				5				5					5
total	5	10	0	0	10	5	10	0	0	15	5	0	5	0	10
total par 4 semaines	5	**50**										5		10	

Synthetic Table of blood sample volumes taken for the trial

Biological samples

The samples that are taken during the trial (serum, PBMC, bone marrow) will be stored in a biological sample bank.

During the trial, the samples will be stored at the CRB Cancer Saint Louis laboratory (laboratoire CRB, 1 avenue Claude Vellefaux, 75010 Paris) under the supervision of the Centre de Ressources Biologiques (CRB) for samples taken in Robert Debre and Saint Louis Hospital and at the local CRB for samples taken in other centers for a duration of 4 years.

At the end of the trial, the samples may be used for further analysis not described in the initial protocol but which may be useful for our investigation of the leukemia in light of developments in scientific knowledge, provided the subject is informed and gives consent, as stated in the information sheet/consent form.

If the samples are kept at the end of the trial, the sample bank will be declared to the relevant minister [and to the director of the competent regional healthcare authority - if the entity is a health establishment] (Article L. 1243-3 CSP).

Type of sample	Quantity	Storage location	Manager of the sample bank	Purpose of the sample bank	Storage period	Outcome (destruction, etc.)
serum	7 samples	Centre de Ressources Biologiques (CRB)	Philippe Manivet	To perform cytokines dosage	4 years	Destruction
PBMC	4 samples	Centre de Ressources Biologiques (CRB)	Philippe Manivet	To perform ancillary biological study	4 years	Destruction
Total bone marrow	3 samples	Centre de Ressources Biologiques (CRB)	Philippe Manivet	To perform ancillary biological study	4 years	Destruction

5.3. Termination and exit rules

5.3.1. Termination of the trial

The trial ends with the last scheduled visit of the last patient included.

In accordance with guideline on follow-up of patients administered with gene therapy medicinal products, patients enrolled in the CAPTiRALL trial who received a second injection of tisegenlecleucel will be the subject of long term follow up (15 years) through the DESCAR-T cohort. The sponsor will receive from LYRA twice a year a specific database extraction and any SAE or suspected SAE linked to the CAPTiRALL trial or the experimental medicine will be declared to ANSM.

5.3.2. Criteria and procedures for prematurely terminating the trial treatment

Events leading to a prematurely termination of trial treatment are listed below. After nivolumab disruption, adverse event and follow-up data will be monitored until 2 years after tisagenlecleucel infusion according to planned visit for patients who ever received at least one dose of nivolumab. Note that for organizational concerns and ethical reasons, investigational center could collect follow-up data provided by the outreach center of patient.
Nivolumab treatment should be permanently discontinued in a given patient for the following:

- Any Grade 2 drug-related uveitis or eye pain or blurred vision that does not respond to topical therapy and does not improve to Grade 1 severity within the re-treatment period OR requires

systemic treatment

- Grade 3 drug-related uveitis, pneumonitis, bronchospasm, neurological event, hypersensitivity reaction, or infusion reaction of any duration requires discontinuation
- Grade 3 drug-related laboratory abnormalities should NOT require permanent discontinuation EXCEPT for total bilirubin > 5 x ULN and/or AST or ALT > 10 x ULN or Concurrent AST or ALT
  > 3 x ULN and total bilirubin > 2 x ULN
- Any other Grade 3 non-skin, drug-related AE lasting > 7 days
- Any Grade 4 nivolumab-related AE or laboratory abnormality, except for the following events which do not require a definitive discontinuation:
  - Grade 4 lymphopenia, leukopenia, anemia, thrombocytopenia
  - Isolated Grade 4 electrolyte imbalances/abnormalities that are not associated with clinical sequelae and are corrected with supplementation/appropriate management within 72 hours of their onset
  - Grade 4 drug-related endocrinopathy AEs such as adrenal insufficiency, ACTH (Adrenocorticotropic Hormone) deficiency, hyper- or hypothyroidisms, or glucose intolerance, which resolve or are adequately controlled with physiologic hormone replacement (steroids, thyroid hormones) or glucose controlling agents, respectively, may not require discontinuation after discussion with the investigator
  - Grade 4 neutropenia persisting 3 months after tisagenlecleucel infusion require discussion with the coordinating investigator
  - Any dosing delay lasting > 6 weeks from the previous dose with the following exceptions: Dosing delays to allow for prolonged steroid tapers to manage drug-related adverse events are allowed. Prior to re-initiating treatment in a subject with a dosing delay lasting > 6 weeks, the coordinating investigator must be consulted.
  - Dosing delays > 6 weeks that occur for non-drug-related reasons may be allowed if approved by the coordinating investigator. Prior to re-initiating treatment in a subject with a dosing delay lasting > 6 weeks, the coordinating investigator must be consulted.

- Any adverse event, laboratory abnormality, or intercurrent illness which, in the judgment of the Investigator, presents a substantial clinical risk to the subject with continued nivolumab dosing.
- Initiation of antineoplastic therapy (ie, chemotherapy, immunotherapy, radiation therapy).
- Disease progression***** and after discussion with the coordinating investigator.
- Pregnancy

*Disease progression is defined as follows:
For Cohort 1:

- reappearance of MRD-confirmed by 2 BM samples at a 2 week-interval
- any type of morphological disease (blood, marrow, CNS)
- clinical sign of CNS involvement (i.e. facial palsy) For Cohort 2:
- if MRD+ at inclusion: increase by at least one log of MRD-confirmed at 2 BM samples at a 2 week-interval
- if morphological relapse at inclusion: M2 or M3 marrow at D28 or later
- any type of morphological disease (blood, marrow, CNS)
- clinical sign of CNS involvement (i.e. facial palsy)

–
- In case of on

- Premature termination of treatment and exit from the study.The investigator must:
  - Document the reason(s)
  - Collect all endpoints at the moment the subject exits from the trial, if the subject agrees
  - Schedule further follow-up visits, especially in case of a serious adverse event. Note that for organizational concerns and ethical reasons, investigational center could collect follow-up data provided by the outreach center of patient.

5.3.3. Criteria and procedure for premature withdrawals and exits from the trial

- Subjects may exit the trial at any time and for any reason.
- The investigator can temporarily or permanently withdraw a subject from the trial for any safety reason or if it is in the subject's best interests.

➔Subject lost to follow-up: the subject cannot be located. The investigator must make every effort to reconnect with the subject (and record his attempts in the source file), at least to determine whether the subject is alive or dead

If a subject exits the trial prematurely or withdraws consent, any data collected prior to the date of premature exit may still be used.

The case report form must list the various reasons why the subject exited or was withdrawn from the trial:

- ☐ Lack of efficacy
- ☐ Adverse reaction
- ☐ Other medical problem
- ☐ Subject's personal reasons
- ☐ Explicit withdrawal of consent
- ☐ Lost to follow-up

5.3.4. Monitoring subjects after the premature termination of treatment

If a subject exits the trial this will in no way affect the standard care received for his/her condition. In case of severe adverse events, the investigator must notify the sponsor and monitor the subject for 12 months following the premature termination of treatment. If treatment is stopped prematurely due to a serious adverse event, a serious adverse event report will be sent by e-mail (eig-vigilance.drc@aphp.fr) to the sponsor. The serious adverse reaction will be monitored until it is resolved.
If a Data Monitoring Committee has been created, the committee can specify and/or validate the monitoring methods.

5.3.5. Full or partial cancellation of the trial

AH-HP (the sponsor) or the Competent Authority (ANSM) may prematurely discontinue all or part of the trial, temporarily or permanently, upon the recommendation of the Data Monitoring Committee in the following situations:

- first, if suspected unexpected serious adverse reactions (SUSARs) are observed or if there is an unexpected increase in the serious adverse reactions, requiring a reassessment of the benefit-risk ratio for the trial.
- similarly, AH-HP, as the sponsor, or the Competent Authority may prematurely cancel the trial due to the discovery of unexpected facts or new information about the product, in light of which the objectives of the trial or clinical program are unlikely to be achieved
- AP-HP, as the sponsor, reserves the right to permanently suspend enrollment at any time if the enrollment targets have not been met.

If the trial is cancelled prematurely, AP-HP will inform the Competent Authority and the Review Board of its decision without undue delay but no later than 15 days of the date of temporary halt or early termination. It shall include the reasons for such action and specify follow-up measure, together with justification for the decision and any recommendations from the Data Safety Monitoring Committee.

6. ELIGIBILITY CRITERIA

The investigator or designee must ensure that only patients who meet all the following inclusion and none of the exclusion criteria are offered enrollment in the trial.

6.1. Observational cohort

All patients, who received a tisagenlecleucel infusion, and who presented an early loss of B-cell aplasia (B cell aplasia is defined by blood B lymphocytes < 10 /mm3 and/or < 3% of total lymphocytes) with or without 2nd tisagenlecleucel product available will be registered in the observational part of the protocol, unless opposition from the parents/child. Patients presenting a B-ALL relapse (either positive MRD or morphological relapse) will be excluded.
Minimal medical data (i.e., baseline data, treatment received, last-disease status, last follow-up) will be collected anonymously from patient records. Non-opposition of patients and or their parents/guardians is warranted before registering.

6.2. Inclusion criteria

- All cohorts
    - Patients aged from 1 to 25 years (pediatric and young adults)
    - Relapsed or refractory CD19+ B-ALL (any relapse after HSCT, 2nd relapse or later, refractory ALL)
    - Life expectancy > 12 weeks.
    - Karnofsky (age > 16) Lansky (age < 16) > 70 at screening
    - No organ dysfunction
    - All patients must have a second tisagenlecleucel (Kymriah®) product available
    - Affiliation to social security or any health insurance (as a beneficiary or assignee).

- Cohort 1:
    - Previously treated by tisagenlecleucel (Kymriah®), and who present an early loss of B-cell aplasia defined by blood B lymphocytes < 10 /mm3 and/or < 3% of total lymphocytes (< 6 months after infusion) while still being in CR with undetectable MRD

- Cohort 2:
    - Previously treated by tisagenlecleucel (Kymriah®), who present a loss of B-cell aplasia defined by blood B lymphocytes < 10 /mm3 and/or < 3% of total lymphocytes and a CD19+ ALL detectable disease in the marrow and/or blood and/or CSF

6.3. Exclusion criteria

- Patient has received systemic therapy* intervening therapy for leukemia after first tisagenlecleucel infusion (chemotherapy, anti-leukemic immunotherapy, ITK, allogeneic HSCT).
  *systemic therapies exclude intrathecal therapy
- Patient has an active auto-immune disease requiring systemic treatment within the past 2 years.
- Patient has known history of, or any evidence of active, non-infectious pneumonitis.
- Patient has a history of non-infectious pneumonitis that required steroid or has current pneumonitis.
- Had receive prior therapy with an anti-PD1, Anti-PDL1 or anti-PDL2 agent.
- Patient has hypersensivity to pembrolizumab/ nivolumab or one of its excipients
- Patient has hypersensivity to Kymriah® or one of its excipients
- Patient has hypersensivity to Fludarabine and/or cyclophosphamide or one of their

excipients

- Patient has received a live vaccine injection within 45 days of planned start of trial therapy.
- Patients with concomitant genetic syndromes associated with bone marrow failure states: such as patients with Fanconi anemia, Kostmann syndrome, Shwachman syndrome or any other known bone marrow failure syndrome. Patients with Down Syndrome will not be excluded.
- Patients with Burkitt's lymphoma/leukemia
- Prior malignancy, except carcinoma *in situ* of the skin or cervix treated with curative intent and with no evidence of active disease.
- Prior treatment with any gene therapy product except first tisagenlecleucel (Kymriah®) injection.
- Prior treatment with any anti-CD19/anti-CD3 therapy, or any other anti-CD19 therapy, except for patients pre-treated with blinatumomab and/or tisagenlecleucel (Kymriah®).
- Prior anti-cancer monoclonal antibody within 4 weeks before starting the trial.
- Prior chemotherapy, targeted small molecule therapy, or radiation therapy within 2 weeks prior to CAR T cells reinfusion or who has not recovered (i.e., ≤ Grade 1 or at baseline) from adverse events due to a previously administered agent.
- Active or latent hepatitis B or active hepatitis C (test within 8 weeks of Screening), or any uncontrolled infection at Screening.
- Human immunodeficiency virus (HIV) positive test within 8 weeks of Screening.
- Presence of grade 2 to 4 acute or extensive chronic GVHD.
- Active CNS involvement by malignancy, defined as CNS-3 per NCCN guidelines. Note: Patients with history of CNS disease that has been effectively treated will be eligible.
- Uncontrolled acute life threatening bacterial, viral or fungal infection at Screening.
- Active infection with COVID19. In case of prior history of COVID19 infection a negative nasopharyngeal test for PCR detection will be required.
- Previous or concurrent malignancy with the following exceptions:
    - Adequately treated basal cell or squamous cell carcinoma (adequate wound healing is required prior to trial entry).
    - *In situ* carcinoma of the cervix or breast, treated curatively and without evidence of recurrence for at least 3 years prior to the trial.
    - A primary malignancy which has been completely resected and in CR for ≥ 5 years
- Pregnant or lactating women (female trial participants of reproductive potential must have a negative serum or urine pregnancy test performed within 48 hours before infusion)

6.4. Recruitment methods

	Number of subjects
Total number of subjects to be included	21(cohort 1) + 5 (cohort 2)
Number of sites	12
Enrollment period (months)	24
Number of subjects/site	5.2
Number of subjects/site/month	**1**

7. TREATMENT ADMINISTERED TO TRIAL PARTICIPANTS

7.1. The investigational medicinal products

7.1.1. Investigational medicinal product 1

- Anti PD1 antibody: nivolumab (Opdivo®) 3 mg/kg every 2 weeks- starts at day +14 or 11 or 5 or -1 after infusion and to be continued until 12 months in case of efficacy.
- BMS already agrees to provide nivolumab (Opdivo®). Treatment could be administered for a maximum of 12 months for each patient.

7.1.2. Investigational medicinal product 2

Tisagenlecleucel (Kymriah®) –additional dose of CAR T cells provided by Novartis. Manufacturing of tisagenlecleucel generally leads to 2 to 3 bags collection. One or 2 bags (max 3 bags) could be delivered for the first infusion to obtain the targeted dose, but the company has already authorized an access to a second bag in the first 2 cases of reinfusion.

Tisagenlecleucel is an autologous cellular immunotherapy product that is comprised of CD3+ T cells that have undergone ex vivo T cell activation, gene modification, expansion and formulation in infusible cryomedia. The transgene to be expressed via lentiviral vector transduction is a CAR targeted against the CD19 antigen. The CAR contains a murine scFv that targets CD19 linked to a transmembrane region derived from the CD8 receptor, which is linked to an intracellular bipartite signaling chain of TCR-ζ (or CD3-ζ) and 4-1BB intracellular signaling domains. The extracellular scFv with specificity for CD19 is derived from a mouse monoclonal antibody. T cells which are enriched from a patient leukapheresis unit are expanded ex vivo using commercially available magnetic beads that are coated with anti-CD3 and anti-CD28 monoclonal antibodies. The cells are transduced with the CD19 CAR lentiviral vector which ensures that only peripheral white blood cells enriched for lymphocytes are exposed to the vector. The residual non-integrated vector is washed away during the process. CTL019 cells expand ex vivo for approximately 10 days. At the end of the culture, the CTL019 cells are depleted of magnetic beads, washed, concentrated, and cryopreserved. Results from a release testing procedure are required prior to release of the product for infusion. A target per-protocol dose of CTL019 transduced cells for paediatric and young adult patients will consist of a single infusion of 0.2 to 5.0 x 10^6 CTL019 transduced viable T cells per kg body weight (for patients ≤ 50 kg) and 0.1 to 2.5 x 10^8 CTL019 transduced viable T cells (for patients
> 50 kg). The following cell dose ranges may be infused if all other safety release criteria are met: 0.2 to 5.0 x 10^6 CTL019 transduced viable T cells per kg body weight (for patient ≤ 50 kg) and 0.1 to 2.5 x 10^8 CTL019 transduced viable T cells (for patients > 50 kg).

7.2. Auxiliary medicinal products

7.2.1. Auxiliary medicinal product 1

Fludarabine (nucleoside analog): 30 mg/m² i.v. daily for 4 doses as part of lymphodepleting chemotherapy

7.2.2. Auxiliary medicinal product 2

Cyclophosphamide (alkylating agent): 500 mg/m² i.v. daily for 2 doses as part of lymphodepleting chemotherapy

7.2.3. Auxiliary medicinal product 3

As rescue treatment after infusion of Kymriah, 2 doses of Tocilizumab must be available
Tocilizumab: 8 mg/kg for patients weighing ≥ 30kg and 12 mg/kg for patients weighing < 30 kg; i.v. infusion over 1 hour.

7.2.4. Auxiliary medicinal product 4

In case of prior hemorragic cystisis during previous treatment by Cyclophosphamide, it is replaced by: Cytarabine 500mg/m²/day from D1 to D2.

7.2.5. Auxiliary medicinal product 5

In case of prior hemorragic cystisis during previous treatment by Cyclophosphamide, it is replaced by Etoposide 150mg/m²/day from D1 to D3.

7.2.6. Auxiliary medicinal product 6

As rescue treatment after infusion of Kymriah, 1 dose of Siltuximab must be available.

siltuximab could be used, IV at 11mg/kg over 1 hour, as recommended in case of refractory CRS to tocilizumab (Santomasso BD., 2021) (Yakoub-Agha I., 2019).

7.2.7. Auxiliary medicinal product 7

As premedication for nivolumab, and Kymriah® infusion: Paracetamol.

7.2.8. Auxiliary medicinal product 8

As premedication for nivolumab, and Kymriah® infusion: polaramine®.

7.3. Traceability information for the investigational medicinal product(s)

Nivolumab is provided by BMS, as commercial form to the DEC-AGEPS, which is in charge of labelling and dispatching to centres.
The sponsor provides each centres with the documents necessary to collect traceability information.
Kymriah® is provided by Novartis directly to the centers. Sponsor provides regulatory labels to the pharmacists involved.

7.4. Authorized and prohibited treatments (medicinal, non-medicinal, surgical), including emergency medications:

7.4.1. Authorized treatment during trial

- Clinically significant prescribed and non-prescribed medications taken by patient within 30 days before inclusion and during the 4 weeks after last nivolumab injection will be recorded in the medical record and in the appropriate CRF.

- Patients included in the trial will receive in addition to nivolumab (Opdivo®) starting on day 14 (or D11, D5 or D-1), a lymphodepleting chemotherapy containing Fludarabine 30mg/m2 intraveinously daily for 4 doses, and cyclophosphamide 500mg/m2 daily IV for 2 doses beginning with the first day of Fludarabine.
- In case of CRS, different drugs could be used:
    - vasopressive drug (like noradrenalin)
    - interleukin 6 drug (like Tocilizumab)
    - Corticosteroids (see in appendix 18.2).
- GM-CSF should be avoided due to the potential of worsen CRS syndrome.
- G-CSF should not be given within 10 days after tisagenlecleucel infusion and until CRS is resolved.
- Corticosteroid therapy should not be used as premedication for tisagenlecleucel infusion or after infusion except for physiological glucosteroid replacement therapy (max 12mg/m²/day hydrocortisone or equivalent).
- In case of immune-related adverse event, use of glucosteroid or anti-metabolite (mycophenolate mofetil) or calcineurine inhibitor should be discussed with principal investigator.

7.4.2. Prohibited treatments

Concurrent use of systemic steroids or immunosuppressant medications are prohibited under this protocol except as required for physiologic replacement of hydrocortisone, or in the case of a life- threatening emergency, since this may have an adverse effect of CAR T- cell expansion and function.

In cohort 1: As patients in cohort 1 must be in MRD negative complete remission, no anti-leukemic concomitant medications are expected. Replacement hydrocortisone and Immunoglobulin substitution are allowed.

In cohort 2: patients should be in molecular or morphological relapse. Thus, the following medications are excluded:

- **Steroids:** Therapeutic systemic doses of steroids must be stopped > 72 hours prior to tisagenlecleucel infusion. However, the following physiological replacement doses of steroids are allowed: < 12 mg/m²/day hydrocortisone or equivalent
- **GVHD therapies:** Any systemic drug used for GVHD must be stopped > 4 weeks prior to tisagenlecleucel infusion to confirm that GVHD recurrence is not observed
- **Chemotherapy:**

- Tyrosine kinase inhibitors and hydroxyurea must be stopped > 96 hours prior to tisagenlecleucel infusion
- The following drugs must be stopped > 1 week prior to tisagenlecleucel infusion and should not be administered concomitantly or following lymphodepleting chemotherapy: vincristine, 6 - mercaptopurine, 6-thioguanine, methotrexate < 25 mg/m2, cytosine arabinoside < 100 mg/m2/day, asparaginase (non-pegylated)

- **CNS disease prophylaxis:**

CNS prophylaxis treatment must be stopped > 1 week prior to tisagenlecleucel infusion (e.g.intrathecal methotrexate)

- **Radiotherapy:**

Non-CNS site of radiation must be completed > 2 weeks prior to tisagenlecleucel infusion CNS directed radiation must be completed > 8 weeks prior to tisagenlecleucel infusion

7.5. Methods for monitoring compliance with the treatment

Both tisagenlecleucel and nivolumab will be administered intravenously in the investigational centers. Compliance to monitoring procedures will be closely recorded.

8. EFFICACY ASSESSMENT

8.1. Description of parameters for assessing efficacy endpoints

8.1.1. Co-Primary end-points

1. limiting-toxicities within the first 28 days after tisagenlecleucel

Limiting toxicities are defined at day 28 after tisagenlecleucel infusion, by the occurrence of either

- Related to CAR-T cells infusion:
  - CRS or aGVH: limiting toxicity will be defined as toxicity ≥ 4
  - ICANs: limiting toxicity will be defined as toxicity ≥ 3 (excepted grade 3 seizure, ie focal, generalized seizure that resolves rapidly)
- Related to nivolumab: immune related myocarditis, pneumonitis, encephalitis: limiting toxicity will be defined as toxicity ≥ 4

2. **Efficacy,** defined at M3 after Tisagenlecleucel reinfusion, by MRD negative CR and B cell aplasia

- Subcohort 1.14: after 6 injections of nivolumab
- Subcohort 1.11: after 6 injections of nivolumab
- Subcohort 1.5: after 7 injections of nivolumab
- Subcohort 1.-1: after 7 injections of nivolumab
- Cohort 2: after 7 injections of nivolumab

MRD negative CR is defined by undetectable disease at a 10^{-4} sensitivity threshold
B cell aplasia is defined by blood B lymphocytes < 10 /mm3 and/or < 3% of total lymphocytes

8.1.2. Secondary end-points

- Incidence of B Cell aplasia at 6 months after infusion of tisagenlecleucel
- Increase of B cell aplasia duration compared to the previous one observed after the first infusion of tisagenlecleucel (up to 24 months).
- Disease best response.
  - CR is defined by less than 5% of blasts in BMA (morphological remission)
  - MRD negative status is defined by undetectable disease at a 10^{-4} sensitivity threshold
  - Disease progression is defined as follows:
    - For Cohort 1
      - Reappearance of MRD-confirmed at 2 BM samples at a 2 week-interval or
      - Any type of morphological disease (blood, marrow, CNS) or
      - Clinical sign of CNS involvement (i.e. facial palsy)
    - For Cohort 2
      - if MRD+ at inclusion: increase by at least one log of MRD-confirmed at 2 BM samples at a 2 week-interval
      - if morphological relapse at inclusion: M2 or M3 marrow at D28 or later
      - Any type of morphological disease (blood, marrow, CNS) or
      - Clinical sign of CNS involvement (i.e., facial palsy)
- Complete remission (measured at M1 M3 M6 M12 after reinfusion)
- Minimal residual disease (measured at M1 M3 M6 M12 after reinfusion)
- 1-year OS
- 1-year EFS
  - Disease response will be assessed each month after infusion for the first 6 months then at M12 if no hematological abnormality
  - Minimal residual disease will be assessed by flow cytometry (if relevant) and/or qPCR at each point of disease response evaluation.

- Incidence of Grade 3 adverse events up to 2 years such as
    - nivolumab-related adverse events: myocarditis, pneumonitis, encephalitis
    - tisagenlecleucel reinfusion-related events, in particular CRS or ICANs or aGVH, prolonged cytopenias
- Incidence of Grade 3, 4 or 5 nivolumab-related adverse events up to 2 years: gut, liver, endocrine, stomatitis, rash or hematologic toxicity
- Incidence of GVHD up to one year

8.1.3. Exploratory end-points

Long-term efficacy measured on
- 2-year OS
- 2-year EFS

To explore the PD1 PDL1 axis and its correlation with success or failure

9. STEERING COMMITTEE FOR THE TRIAL

- Members of the committee: Andre Baruchel, Sylvie Chevret, Nicolas Boissel, Audrey Grain, Marie Emilie Dourthe
- Missions: oversight of the protocol opening and progression (validation of limiting toxicities)
- Operating procedures: every other week meeting (visio/phone/physical as needed and possible) and more frequent if necessary

10. SAFETY ASSESSMENT - RISKS AND RESTRICTIONS ADDED BY THE Trial

10.1. Recording and reporting adverse events

10.1.1. Definitions

According to Article 2 of the Regulation (EU) N° 536/2014:

- **Adverse event**

  Any untoward medical occurrence in a trial subject, to whom a medicinal product is administered and which does not necessarily have a causal relationship with the treatment.

- **Serious adverse event**

  Any untoward medical occurrence that at any dose requires inpatient hospitalisation or prolongation of existing hospitalisation, results in persistent or significant disability or incapacity, results in a congenital anomaly or birth defect, is life-threatening, or results in death.

- **Unexpected serious adverse reaction to an investigational medicinal product**

  A serious adverse reaction, the nature, severity or outcome of which is not consistent with the reference safety information.

According to Article 53 of the Regulation (EU) No 536/2014:

- **Unexpected event**

An unexpected event, which affect the benefit-risk balance of the clinical trial, but are not suspected unexpected serious adverse reactions as referred to in Article 42. That notification shall be made without undue delay but no later than 15 days from the date the sponsor became aware of this event.

According to Article 54 of the Regulation (EU) No 536/2014:

- **Urgent safety measure**

Where an unexpected event is likely to seriously affect the benefit-risk balance, the sponsor and

the investigator shall take appropriate urgent safety measures to protect the subjects.
The sponsor shall notify the Member States concerned, through the EU portal, of the event and the measures taken.
That notification shall be made without undue delay but no later than seven days from the date the measures have been taken.

10.1.2. The role of the investigator

The investigator must **assess the seriousness criteria of each adverse event** and record all serious and non-serious adverse events in the case report form (eCRF).
The investigator must **document** serious adverse events **as thorough as possible** and provide a definitive medical diagnosis, if possible.

The investigator must **assess the severity** of the adverse events by using:
- either general terms:
    - *Mild: tolerated by the patient, does not interfere with daily activities*
    - *Moderate: sufficiently uncomfortable to affect daily activities*
    - *Serious: preventing daily activities*
- or a severity grading scale for adverse events, attached to the protocol
    - Common Terminology Criteria for Adverse Events [National Cancer Institute]

The investigator must assess the **causal relationship** between the serious adverse events and the investigational medicinal product(s).

The method used by the investigator is based on the 2 causality terms (EVCTM method:
- Related
- Not related
- Possible
- Unlikely (not ruled out).

10.1.3. Serious adverse events that require a notification without delay by the investigator to the sponsor

, the investigator must notify the sponsor **without undue delay but not later than within 24 hours on the day when the investigator becomes aware** of any serious adverse event which occurs during a trial as described in Article 41of Regulation (EU) N°536/2014, except those which are listed in the protocol and, if applicable, in the investigator's brochure as not requiring a notification without delay.
A serious adverse event is any untoward medical occurrence that:

1 results in death
2- is life-threatening
3- requires inpatient hospitalization or prolongation of existing hospitalization 4- results in persistent or significant disability/incapacity
5- is a congenital anomaly/birth defect
6- Adverse event deemed “medically significant”

Medication errors, pregnancies and uses outside what is foreseen in the protocol, including misuse and abuse of the product, require the investigator to notify the sponsor without delay.

An AE fulfilling the SAE-criteria will not be reported on an SAE-form if it falls into a category of exceedingly common expected toxicities for which there is no specific interest in the protocol evaluation, listed under Exception rules for SAE-reporting in part **10.1.2.2.2**

This means that for the majority of the protocol, SAE-reporting within 24 hours is limited to toxicities fulfilling the SAE-criteria but not fulfilling the criteria for AESM and not being listed as exempt from expedited SAE-reporting.

10.1.4. Specific features of the protocol

10.1.4.1. Other events that require the investigator to notify the sponsor without delay

- Adverse events judged as being "medically significant"

defined by an important medical event that may jeopardize the subject or may require intervention (eg, medical, surgical) to prevent one of the other serious outcomes listed in the definition above.
Examples of such events include, but are not limited to, intensive treatment in an emergency room or at home for allergic bronchospasm; blood dyscrasias or convulsions that do not result in hospitalization.)

The investigator must notify the sponsor without delay on the day the investigator become aware of these adverse events, in the same manner and within the same deadline as for serious adverse events (see above).

- In utero exposure

The investigator must notify the sponsor without delay on the day the investigator becomes aware of any pregnancy that occurs during the trial, even if not associated with an adverse event.
If the investigational medicinal product is genotoxic, every case of maternal or paternal exposure must be notified.
Notify using a specific form appended to the protocol.

- Exposure through breastfeeding

Exposure through breastfeeding occurs if an infant or a child could have been exposed *via* the breast milk of a mother being treated with an investigational medicinal product.
Even if it is not associated with an adverse event, the investigator must notify the sponsor without delay on the day the investigator becomes aware about the exposure while breastfeeding

10.1.4.2. Serious adverse events that do not require the investigator to notify the sponsor without delay *(Exception to SAE rules)*

These serious adverse events are simply recorded in the case report form.

- Normal and natural course of the condition:
  - Planned hospitalization for monitoring the condition under investigation no deterioration in the participant's condition compared to baseline],
  - hospitalization for routine treatment or for monitoring of the condition under investigation, not associated with a deterioration in the participant's condition, emergency hospitalization at inclusion or prolongation of hospitalization after inclusion for monitoring the condition under investigation

- *Special circumstances*
  - Hospitalization for a pre-existing illness or condition
  - Hospitalization for a medical or surgical treatment scheduled arranged prior to the trial
  - Admission for social or administrative reasons

- *Adverse events potentially related to treatments prescribed as part of the care provided during the trial follow-up*

The investigator must report these adverse events to the relevant regional pharmacovigilance centre (Centre Regional de Pharmacovigilance CRPV).

An AE fulfilling the SAE-criteria will not be reported on an SAE-form if it falls into a category of exceedingly common expected toxicities for which there is no specific interest in the protocol evaluation, listed below. These specific SAE will be extracted by the clinical research unit to be transmitted to the relevant pharmacovigilance center every 6 months.

This means that for the majority of the protocol, SAE-reporting within 24 hours is limited to toxicities fulfilling the SAE-criteria but not fulfilling the criteria for adverse events specially followed by the promoter and not being listed as exempt from expedited SAE-reporting.

Because of the extensive expected toxicity, full reporting of adverse events according to standard guidelines is of limited value when evaluating toxicity in an ALL trial.
Thus, it is not necessary to report SAEs that only require hospitalization or prolonged hospitalization, because the intensity of the therapy will lead to too many SAEs by this definition. The following anticipated adverse events (until grade 3 included) (see the list below) are excluded from the SAE reporting requirements.

- Febrile neutropenia with the need for inpatient antibiotic therapy
- Infection without neutropenia requiring in-patient antibiotic therapy
- Allergic reactions
- Osteonecrosis
- Cushing syndrome Diabetes mellitus
- Tissue necrosis after paravasal injection
- Cystitis
- Rash
- Diarrhea, nausea, vomiting
- Impaired liver-function or elevated liver enzymes
- Hospitalizations planned before entry into the clinical trial
- Elective treatment of a pre-existing condition
- Post lumbar puncture syndrome
- Catheter-related incidents

10.1.4.3. Adverse Event specially followed by the promotor

The sponsor should closely monitor these specific AE.
These adverse events are defined by immune and non-immune related adverse events i.e., linked to nivolumab (Opdivo®) treatment such as: gut, liver, lung, endocrine effect, pneumonitis, stomatitis, rash.
Neurotoxicity and CRS linked to tisagenlecleucel (Kymriah®) infusion.

Siltuximab is the subject of specific attention by ANSM. Data from patients who received Siltuximab as treatment for CRS will be collected and related AE with will be closely monitored.

These adverse events will be submitted periodically to DSMB members and the DSMB minute will be transmitted to Competent Authority.

It should be reported to the sponsor within 72 hours using the dedicated report form. The sponsor will review all these adverse events every 6 months until the end of the trial. In case of serious adverse event the investigator most notify the sponsor **without delay on the day when the investigator becomes aware.**

10.1.4.4. Adverse events during the trial possibly related with the treatments prescribed as part of the patient's standard care

The investigator must report these events to his *Centre Régional de Pharmacovigilance* (CRPV).

10.1.4.5. Period during which the investigator must send notification of SAEs to the sponsor without delay

The investigator notifies the sponsor without delay of all the serious adverse events listed in the corresponding section:

- starting from the date on which the subject undergoes the first intervention specific to the trial
- throughout the whole follow-up period intended by the trial
- until 4 weeks or more after the end of the subject's treatment with the investigational medicinal product. For nivolumab, related SAE have to be followed until 100 days after the last infusion.
- indefinitely, if the SAE is likely to be due to the investigational medicinal product or to the trial interventions (e.g., serious reactions that could appear at long term after exposure to the medication, such as cancers or congenital abnormalities)

10.1.4.6. Procedures and deadlines for notifying the sponsor

The investigator should initially complete a SAE reporting form (contained in the case report form). This report must be signed by the investigator.

The investigator must complete every section of the SAE form so that the sponsor can carry out the appropriate assessment.

The initial report sent to the sponsor must be rapidly followed up by one or more additional written reports describing the course of the event and any complementary information.

Whenever possible, the investigator will provide the sponsor with any documents that may be useful for medical assessment of the case (medical reports, laboratory test results, results of additional exams, etc.). These documents must be anonymized. In addition, the investigator must state the trial acronym and the number and initials of the trial participant on each paper.

Any adverse event will be monitored until fully resolved (stabilization at a level considered acceptable by the investigator, or return to the previous state) even if the subject has terminated his participation in the trial.

The initial report, the SAE follow-up reports and all other documents must be sent to the sponsor's safety Department by e-mail at eig-vigilance.drc@aphp.fr. It should be noted that it is possible to send SAE reports to the Safety Department by fax to +33 (0)1 44 84 17 99, only in the event of a failed attempt to send the SAE report by email (in order to avoid duplication).

For trials which use e-CRF

- the investigator completes the SAE report form in the e-CRF, then validates, prints and signs the form before sending it by fax;
- In case of failure to connect to the e-CRF, the investigator should complete, sign and send the SAE report form to the safety Department. As soon as the connection is restored, the investigator must complete the SAE report form in the e-CRF.

The investigator must comply with all requests for additional information from the sponsor.
For all questions relating to an adverse event report, the safety Department can be contacted via email at vigilance.drc@aphp.fr.

10.1.5. Role of the sponsor

The sponsor, represented by its safety Department, shall continuously assess the safety of each investigational medicinal product throughout the trial.

Serious adverse events related to nivolumab will be reported to marketing authorization holder within 24 hours (1 business day) of becoming aware of the event.

10.1.6. Analysis and declaration of serious adverse events

The sponsor assesses:

- the **seriousness** of all reported adverse events,
- the **causal relationship** between these adverse events and investigational medicinal

product and any other treatments,
All serious adverse events for which the investigator and/or the sponsor suspect a causal relationship with the investigational medicinal product are classed as suspected serious adverse reactions.

- the **expectedness assessment** of the serious adverse reactions
Any serious adverse reaction whose nature, severity, frequency or outcome is inconsistent with the safety information described in the summary of product characteristics, or in the investigator's brochure if the product is not authorized, is considered unexpected.
The sponsor, acting through its safety Department, assesses the expectedness of the serious adverse reaction based on the information described below**.**

Any suspected unexpected serious adverse reaction must also be declared electronically using the Eudravigilance European adverse drug reactions database managed by the European Medicines Agency (EMA).

The sponsor must notify all the investigators about any information that could adversely affect the safety of the trial subjects.

❖ For serious adverse events likely to be related to the investigational medicinal products:
- Tisagenlecleucel (Kymriah®), refer to the SMPC enclosed in CTIS platform.
- Nivolumab (Opdivo®), refer to the Investigator's Brochure enclosed in CTIS platform.

❖ For the serious adverse events that may be related to the Auxiliary medicinal products:
- Fludarabine refer to the SmPC enclosed in CTIS platform
- Cyclophosphamide refer to the SmPC enclosed in CTIS platform
- Uromitexan (Mesna®) refers to the SMPC enclosed in CTIS platform
- Tocilizumab refer to the SmPC enclosed in CTIS platform
- Cytarabine refer to the SmPC enclosed in CTIS platform
- Etoposide refer to the SmPC enclosed in CTIS platform
- Siltuximab refer to the SmPC enclosed in CTIS platform
- As premedication for nivolumab, and Kymriah® infusion: paracetamol refer to the SmPC enclosed in CTIS platform
- As premedication for nivolumab, and Kymriah® infusion: dexchlorpheniramine (polaramine®) refer to SMPC enclosed in CTIS platform

❖ The serious adverse events potentially related to the interventions, procedures or examinations specific to the trial are:
- Expected SAE related to myelogramme puncture: pain, bleeding, redness or swelling
- Expected SAE related to lumbar puncture: headache, low back pain, nausea and vomiting, dizziness, irritation of a nerve causing temporary numbness, tingling or pain in legs, infection where the needle was inserted, bleeding in the spinal canal
- Biological samples: Not applicable

The sponsor will report all suspected unexpected serious adverse reactions (SUSARs), via Eudravigilance within the regulatory time frame, to the Competent authority:
- in the case of fatal or life-threatening suspected unexpected serious adverse reactions, as soon as possible and in any event not later than seven days after the sponsor became aware of the reaction
- In the case of non-fatal or non-life threatening suspected unexpected serious adverse reaction, no later than 15 days after the sponsor became aware of the reaction;
- In the case of a suspected unexpected serious adverse reaction which was initially considered to be non-fatal or non-life threatening but which turns out to be fatal or life-threatening, as soon as possible and in any event no later than seven days after the sponsor became aware of the reaction being fatal or life-threatening

- In the uncompleted case of fatal or life-threatening suspected unexpected serious adverse reactions, all additional information to complete this case have to be reported as soon as possible and in any event no later than eight days after the initial report
- The sponsor must provide all relevant additional information by sending follow-up reports, within 15 days from learning of the information.

The sponsor must notify all the investigators involved about any information that could adversely affect the safety of the research participants.

10.1.7. Analysis and declaration of Adverse events specially followed by the promotor

Adverse event specially followed by the promotor should be notified to the sponsor within 24 hours, using the dedicated report form. The sponsor assesses every 6 months seriousness of all these AE reported, the causal relationship between these adverse events and the investigational product (Nivolumab). Any suspected unexpected serious adverse reaction must also be declared electronically using the Eudravigilance European adverse drug reactions database managed by the European Medicines Agency (EMA).

The sponsor must notify all the investigators about any information that could adversely affect the safety of the trial subjects.

10.1.8. Analysis and declaration of other safety data

This relates to any new safety data that may lead to a reassessment of the risk/benefit ratio of the trial or the investigational medicinal product, modifications in the investigational medicinal product use, the conduct of the clinical trial, or the clinical trial documents, or a suspension, interruption or modification of the protocol of the clinical trial or other similar trials.
The sponsor will report in CTIS platform and to the competent authority without delay after becoming aware of the emerging safety issue and, if applicable, describe which measures have been taken.

10.1.9. Annual safety report

The sponsor must prepare once yearly throughout the trial duration an annual safety report (Development Safety Update Report - DSUR) which includes, in particular:
- an analysis of safety data concerning trial subjects
- a description of the patients included in the trial (demographic profile etc.)
- a list of all the suspected serious adverse reactions that occurred during the period covered by the report,
- cumulative summary tabulation of all the serious adverse events that have occurred since the beginning of the clinical trial,

The report must be transmitted in CTIS platform no later than 60 days after the anniversary date corresponding to the date of authorization of the clinical trial by Competent Authority.

10.1.10. Data Safety Monitoring Board (DSMB)

A Data Safety Monitoring Board (DSMB) will be set up for this trial. The DSMB must hold its first meeting before the first subject is enrolled. The DSMB's preliminary meeting should take place before the protocol submission to competent health authority (ANSM) and Ethics committee.
Therefore, we scheduled that, after data interim analysis of phase I data once the first 3 children have been enrolled and their DLT assessed – and validated by the trial Steering Committee-, the statistical report will be submitted to the DSMB to obtain its advice regarding the "nivolumab starting day" recommended for further enrolled children.
The role of DSMB overpasses the evaluation of DLTs used in dose-escalation algorithm. We thus schedule that the DSMB meeting occurs at pre-specified time intervals, e.g. every 3 or 6 months. This rhythm would be adapted according to the rhythm of inclusions. Obviously in case of SUSAR occurrence, or grade 5 adverse event, extra meetings will be set up.

This strategy will be presented to the first DSMB meeting, before trial inclusions.

11. DATA MANAGEMENT

11.1. Right to access source data and documents

11.1.1. Access to data

In accordance with GCP:
- the sponsor is responsible for ensuring all parties involved in the trial agree to guarantee direct access to all locations where the trial will be carried out, the source data, the source documents and the reports, for the purposes of the sponsor's quality control and audit procedures.
- the investigators will ensure the persons in charge of monitoring and auditing the clinical trial and of quality control have access to the documents and personal data strictly necessary for these tasks, in accordance with the statutory and regulatory provisions in force (Articles L.1121-3 and R.5121-13 of the French Public Health Code)

11.1.2. Source documents

The source documents are any original document or item that proves the existence or accuracy of a data-point or fact recorded during the trial. Source documents will be kept by the investigator, or by the hospital in the case of hospital medical records, for the statutory period.

11.1.3. Data confidentiality

The persons responsible for the quality control of clinical trials will take all necessary precautions to ensure the confidentiality of information relating to the investigational medicinal products, the trial, the trial participants and in particular the identity of the participants and the results obtained. These persons, as well as the investigators themselves, are bound by professional secrecy. During and after the clinical trial, all data collected about the trial participants and sent to the sponsor by the investigators (or any other specialised collaborators) will be anonymised.
Under no circumstances will the names and addresses of the subjects be shown.
The sponsor will ensure that each subject has agreed in writing for any personal information about him or her which is strictly necessary for the quality control of the trial to be accessed.

11.2. Data processing and storage of documents and data

11.2.1. Identification of the data processing manager and the location(s)

Data management and analyses will be performed at the Centre de traitement de données (CTD) INCA de l'APHP, Service de biostatistique et information médicale, hôpital Saint- Louis, Paris.

11.2.2. Data entry

Data will be entered electronically via a web browser.

11.2.3. Data processing (CNIL, the French Data Protection Authority) in France

This trial is governed by the CNIL "Reference Method for processing personal data for clinical studies" (MR-001, amended). AP-HP, the trial sponsor, has signed a declaration of compliance with this "Reference Method"

11.2.4. Archiving

The specific documents for a clinical trial on a medicinal product for human use will be archived by the investigator and the sponsor for 15 years after the end of the trial.

- This indexed archiving includes, in particular: A sealed envelope containing the originals of

all information sheets and consent forms signed by all individuals at the site who participated in the trial for the investigator;

- One copy of all the information sheets and signed consent forms signed for all individuals at the site who participated in the trial for the sponsor;
- "trial" binders for the Investigator and the sponsor, containing:
  - all successive versions of the protocol (identified by version no. and date), and its appendices
  - the ANSM authorizations and CPP decisions
  - correspondence
  - the enrollment list or register
  - the appendices specific to the trial
  - the final trial report
- The data collection documents

11.2.5. Ownership of the data

AP-HP is the owner of the data. The data cannot be used or disclosed to a third party without its prior permission.

12. STATISTICAL ASPECTS

12.1. Planned statistical methods, including the timetable for any planned interim analyses

The analysis will be performed based on a modified intent-to-treat principle: All participants will be analyzed, regardless of whether they completed the trial, unless they did not receive the reinfusion of tisagenlecleucel.

The baseline characteristics of the patients will be reported, based on point estimations, with 95% confidence intervals.

The analysis of the limiting-toxicity endpoint will use a Bayesian framework.
Sequential analyses of cohort 1 data will be performed after each sub-cohort of 3 patients, following the Time to Event Continual Reassessment Method (TITE-CRM) dose-finding clinical trial design. It is a model-based design, where the dose–limiting toxicity relationship is modeled through a cumulative function corresponding to a one-parameter power model, the parameter of which is sequentially estimated from the data

$$P(Y = 1 \mid x_i) = \psi(x_i, \theta) = (p_i)^{\theta}$$

The estimation will be based on a weighted likelihood introduced by Cheung and Chappell (2000) who observed that each patient may provide a weight, wi, reflecting the extent to which their outcome has been evaluated.

Secondary endpoints will be estimated by maximum likelihood method with 95% confidence intervals
The binary endpoint (best response, complete remission and minimal residual disease at M1 M3 M6 M12) will be estimated with exact 95% confidence interval.
The right-censored endpoints (event-free survival and overall survival) will be estimated by the nonparametric Kaplan-Meier method. Competing risks endpoints (relapse incidence, etc.) will be analyzed using competing risks methods. Specifically, the cumulative incidences of these events will be estimated considering deaths prior to the event of interest as competing risk outcomes.
Grade 3 or 4 adverse events incidence will use the Andersen and Gill model for recurrent events.

All analyses will be performed using the R (http://www.R-project.org/) software packages. The titecrm package will be used for TITE-CRM model-fitting and analysis.

12.2. Hypotheses for calculating the required number of subjects, and the result

In MRD negative patients: The TITE-CRM design has been shown to accurately estimate the optimal schedule based on sample sizes of 15-25 patients. A sample of 21 patients will be enrolled.
In relapsed patients: The Pilot trial will include a maximum of 5 patients

12.3. References

Cheung, Ying Kuen. 2011. Dose Finding by the Continual Reassessment Method. New York: Chapman & Hall / CRC Press.

Cheung, Y K, and R Chappell. 2000. “Sequential Designs for Phase I Clinical Trials with Late-Onset Toxicities.” Biometrics 56 (4): 1177–82.

12.4. State whether subjects who exit the trial prematurely will be replaced and in what proportion.

No patient will be replaced unless consent withdrawal before measurement of the primary endpoint, that avoids the inclusion of the patient in the analysis.

12.5. Statistical criteria for termination of the trial.

No statistical criteria will be defined.

12.6. Method for taking into account missing, unused or invalid data

Given the expected sample size, it is likely that no missing data will be observed. Otherwise, simple imputation, based on the modal class or median value of the parameter, according to the type of variable on the observed data will be performed, with sensitivity analyses to the imputation method.

12.7. Management of modifications made to the analysis plan for the initial strategy.

All modifications will be submitted for approval to the CPP.

12.8. Selection of populations

Primary analyses will be performed on an intent-to-treat basis.
Secondary exploratory analyses will consider the population of compliers, that is, those who completed the treatment according to the scheduled protocol.

13. QUALITY CONTROL AND ASSURANCE

Every clinical trial managed by AP-HP is ranked according to the projected risk incurred by the trial participants using a classification system specific to AP-HP-sponsored clinical trials.

13.1. General organisation

The sponsor must ensure the safety and respect of individuals who have agreed to participate in the trial. The sponsor must have a quality assurance system for monitoring the implementation of the trial at the research centres.

For this purpose, the sponsor shall appoint Clinical Research Associates (CRA) whose primary role is to carry out regular follow-up visits at the trial sites, after completing their initial visits.
The purpose of monitoring the trial, as defined in the Good Clinical Practices (GCP section 5.18.1), is to verify that:

- the research subjects are safe, protected and their rights are being met
- the data being recorded is accurate, complete and consistent with the source documents
- the trial is carried out in accordance with the current version of the protocol, with GCP and with all statutory and regulatory requirements.

13.2. Strategy for site opening

The strategy for opening the sites is determined using the tailored monitoring plan.

13.3. Scope of site monitoring

For this trial, the appropriate monitoring level has been determined based on the complexity, the impact and the budget for the trial. Therefore, the sponsor, in agreement with the coordinating investigator, has agreed on a logistical score and impact and the corresponding trial monitoring level of: **D** level.

13.4. Quality control

A Clinical Research Associate (CRA) appointed by the sponsor will be responsible for the proper running of the trial, for collecting, documenting, recording and reporting all handwritten data, in accordance with the Standard Operating Procedures applied within the DRCD and in accordance with Good Clinical Practice as well as with the statutory and regulatory requirements.
The investigator and the members of the investigator's team agree to make themselves available during regular Quality Control visits by the Clinical Research Associate. During these visits, the following elements will be reviewed:

- written consent
- compliance with the trial protocol and its procedures
- quality of the data collected in the case report forms: accuracy, missing data, consistency of the data with the "source" documents (medical files, appointment books, original copies of laboratory results, etc.)
- management of the treatments used

13.5. Case Report Form

All information required by the protocol must be entered in the case report forms. The data must be collected as and when it is obtained, and clearly recorded in these case report forms. Any missing data must be coded.
Every site will have access to the electronic case report forms via a web-based data collection system. Investigators will be given instructions for using this tool.
Using on-line case report forms means the CRA can view the data quickly and remotely. The investigator is responsible for the accuracy, quality and relevance of all the data entered. In addition, there are consistency checks to ensure the data are verified immediately upon being

entered. The investigator must validate any changes to the values in the case report form. An audit trail will be kept of all changes. A justification can be added when applicable, as a comment. A print-out, authenticated (signed and dated) by the investigator, will be requested at the end of the

trial. The investigator must archive a copy of the authenticated document that was delivered to the sponsor.

13.6. Management of non-compliances

Any events that occur as a result the investigator or any other individual involved in conducting the trial failing to comply with the protocol, standard operating procedures, good clinical practice or statutory and regulatory requirements must be recorded in a declaration of non-compliance and sent to the sponsor.

The sponsor has its own procedures for managing these non-compliances.

13.7. Audits/inspections

The investigators agree to accept the quality assurance audits carried out by the sponsor as well as the inspections carried out by the competent authorities. All data, documents and reports may be subject to regulatory audits and inspections. These audits and inspections cannot be refused on the basis of medical secrecy.

An audit can be carried out at any time by independent individuals appointed by the sponsor. The aim of the audits is to ensure the quality of the trial, the validity of the results and compliance with the legislation and regulations in force.
The persons who manage and monitor the trial agree to comply with the sponsor's requirements and with the competent authority regarding trial audits or inspections.
The audit may encompass all stages of the trial, from the development of the protocol to the publication of the results and the storage of the data used or produced as part of the trial.

13.8. Principal Investigator's declaration of responsibility

Before starting the trial, each investigator will give the sponsor’s representative a signed and dated copy of his/her curriculum vitæ and RPPS number (Répertoire Partagé des Professionnels de Santé, Collective Database of Health Professionals).
Each investigator will agree to comply with legislation and to conduct the trial in line with GCP, in accordance with the Declaration of Helsinki.
The Principal Investigator at each participating site will sign a declaration of responsibility (standard DRCD document) which will be sent to the sponsor's representative.
The investigators and their co-workers will sign a delegation form specifying each person's role.

14. ETHICAL AND LEGAL CONSIDERATIONS

14.1. Methods for informing and obtaining consent from the research participants

In accordance with Article L.1122-2 of the French Public Health Code, when a trial involves a non- emancipated minor, consent must be given by the legal guardians.
The parents or legal guardians will be granted a fair period of reflection before signing the consent form after getting informed about disease and trial.
The free and informed written consent of the legal guardians is obtained by the investigator, or by a doctor representing the investigator, before the minor is enrolled on the trial.

However, in accordance with Article L1122-2 of the French Public Health Code, consent may be given by the only present legal guardians, subject to the following conditions:

- the trial involves only minimal risks and constraints
- the minor is not taking part in the trial as a healthy volunteer

- the other legal guardian is unable to give consent within a suitable deadline given the specific methodology of the trial and its purpose.

Minors will be given the information specified in Article L. 1122-1 of the French Public Health Code, adapted to suit their level of understanding, by both the investigator and their legal guardians. Minors will be asked to agree to take part in the human research trial. In all cases, the investigator must accept a minor's refusal to participate or a withdrawal of their agreement.
A copy of the signed and dated consent form is given to the parents or legal guardians as well as to the investigator or the doctor representing the investigator. The investigator will retain the original.
At the end of the trial, a copy will be placed in a tamper-proof sealed envelope containing all the consent forms. This envelope will be archived by the sponsor.
Minors who reach the age of majority during their participation in the trial will be given the relevant information at that time. After they have been given this information, they will be asked to confirm their consent.

14.2. Legal obligations

14.2.1. The sponsor's role

Assistance Publique Hôpitaux de Paris (AP-HP) is the sponsor of this trial and has delegated powers to its Clinical Research and Development Department (DRCD) in order to conduct the trial in accordance with Article L.1121-1 of the French Public Health Code. AP-HP reserves the right to terminate the trial at any time for medical or administrative reasons. In this case, the investigator will be informed accordingly.

14.2.2. Request for approval from the Institutional Review Board

AP-HP, as sponsor, obtains prior approval from the Institutional s Review Board for its clinical trials of medicinal products for human use, within the scope of the Board's authority and in accordance with statutory and regulatory requirements.

14.2.3. Request for approval from the ANSM

AP-HP, as sponsor, obtains prior authorization from the ANSM for its clinical trials of medicinal products for human use, within the scope of the ANSM's authority and in accordance with statutory and regulatory requirements.

14.2.4. Declaration of compliance with the MR 001 "Reference Method"

- Commitment to comply with “Reference Methodology” MR-001

This research is governed by the CNIL (French Data Protection Agency) “Reference Methodology for processing personal data used within the scope of health research” (amended MR-001).

AP-HP, the trial sponsor, has signed a declaration of compliance with this "Reference Method".

14.2.5. Modifications to the trial

Any substantial amendment made to the protocol by the coordinating investigator must be sent to the sponsor for approval. After approval is given, the sponsor must obtain, prior to implementing the amendment, approval from the Institutional Review Board and authorization from the ANSM, within the scope of their respective authorities.
The information sheet and the consent form can be revised if necessary, in particular if there is substantial amendment to the trial or if adverse reactions occur.

14.2.6. Final trial report

The final trial report referred to in CSP Article R.1123-67 is written and signed by the sponsor

and the investigator. A report summary, meeting the competent authority's guidelines, has to be sent to the competent authority and Institutional Review Board within one year of the end of the trial i.e. the end of the participation of the last trial participant.

15. FUNDING AND INSURANCE

15.1. Sources of funding for the trial

PHRC

15.2. Insurance

For the duration of the trial, the Sponsor will take out an insurance policy covering the sponsor's own third party liability as well as the third party liability of all the doctors involved in the trial. The sponsor will also provide full compensation for any damages caused by the trial to the trial participant and their beneficiaries, unless the sponsor can prove that the harm is not the fault of the sponsor or any collaborator. Compensation cannot be refused on the grounds of a third party act or the voluntary withdrawal of the person who initially consented to participate in the trial.

Assistance Publique-Hôpitaux de Paris (AP-HP) has taken out insurance with HDI-GERLING through BIOMEDIC-INSURE, covering its own third party liability and that of any collaborator (doctor or research staff), in accordance with Article L.1121-10 of CSP.

16. PUBLICATION

"The research was funded by a grant from Programme Hospitalier de Recherche Clinique - PHRC 2010 (Ministère de la Santé)
This research has been registered on the website http://clinicaltrials.gov/ under NCT05310591 registration number.

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18. LIST OF ADDENDA

18.1. List of Investigators

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18.2. CRS grading and treatment recommendations

18.2.1. CRS grading scale according to ASTCT consensus

ASTCT CRS Consensus Grading

CRS Parameter	Grade 1	Grade 2	Grade 3	Grade 4
Fever*	Temperature ≥38°C	Temperature ≥38°C	Temperature ≥38°C	Temperature ≥38°C
		With		
Hypotension	None	Not requiring vasopressors	Requiring a vasopressor with or without vasopressin	Requiring multiple vasopressors (excluding vasopressin)
		And/or†		
Hypoxia	None	Requiring low-flow nasal cannula‡ or blow-by	Requiring high-flow nasal cannula‡, facemask, nonrebreather mask, or Venturi mask	Requiring positive pressure (eg, CPAP, BiPAP, intubation and mechanical ventilation)

Organ toxicities associated with CRS may be graded according to CTCAE v5.0 but they do not influence CRS grading.

* Fever is defined as temperature ≥38°C not attributable to any other cause. In patients who have CRS then receive antipyretic or anticytokine therapy such as tocilizumab or steroids, fever is no longer required to grade subsequent CRS severity. In this case, CRS grading is driven by hypotension and/or hypoxia.

† CRS grade is determined by the more severe event: hypotension or hypoxia not attributable to any other cause. For example, a patient with temperature of 39.5°C, hypotension requiring 1 vasopressor, and hypoxia requiring low-flow nasal cannula is classified as grade 3 CRS.

‡ Low-flow nasal cannula is defined as oxygen delivered at ≤6 L/minute. Low flow also includes blow-by oxygen delivery, sometimes used in pediatrics. High-flow nasal cannula is defined as oxygen delivered at >6 L/minute.

18.2.2. CRS management guidelines according to ELIANA trial

Pretreatment
• Acetaminophen/paracetamol and diphenhydramine/H1 antihistamine • Prophylaxis for complications of TLS as appropriate
Tisagenlecleucel Infusion **Prodromal Syndrome:** low-grade fevers, fatigue, anorexia (hours to days)
• Observation, rule out infection (surveillance cultures) • Antibiotics per local guidelines (febrile neutropenia) • Symptomatic support
Symptom Progression: High fevers, hypoxia, mild hypotension First-Line Management: • Oxygen, fluids, low-dose vasopressor support, antipyretics • Monitor/manage complications of TLS
Further Symptom Progression: • Hemodynamic instability despite intravenous fluids and moderate to "high-dose" vasopressor[1] support *or* • Worsening respiratory distress, including pulmonary infiltrates increasing oxygen requirements including high-flow oxygen and/or need for mechanical ventilation *or* • Rapid clinical deterioration Second-Line Management: Tocilizumab: IV Infusion Over 1 Hour • Patient weight <30 kg: 12 mg/kg i.v. • Patient weight ≥30 kg: 8 mg/kg i.v. (maximum dose, 800 mg) Hemodynamic and respiratory support
Lack of Clinical Improvement While Awaiting Tocilizumab Response Third-Line Management: • Consider other diagnosis causing clinical deterioration (i.e., sepsis, adrenal insufficiency) • If no improvement with first dose of tocilizumab within 12 to 18 hours, consider steroids (plan rapid taper **after** hemodynamic normalization):   ○ 2 mg/kg methylprednisolone as an initial dose, then 2 mg/kg per day. Because steroids are tapered quickly, monitor for adrenal insufficiency and need for hydrocortisone replacement • If no response to steroids within 24 hours, consider second dose of tocilizumab (dosed as above) • Hemodynamic and respiratory support
Lack of Clinical Improvement While Awaiting Response to Third-Line Management Fourth-Line Management: • Consider other diagnosis causing clinical deterioration (i.e., sepsis, adrenal insufficiency) • If no response to steroids and second dose of tocilizumab within 24 hours or further clinical deterioration, consider siltuximab 11 mg/kg IV over 1 hour • Hemodynamic and respiratory support
Lack of Clinical Improvement While Awaiting Response to Fourth-Line Management Fifth-Line Management: • Consider other diagnosis causing clinical deterioration (i.e., sepsis, adrenal insufficiency) • In ongoing CRS despite prior therapy, consider anti–T-cell therapies such as cyclophosphamide, antithymocyte globulin, or alemtuzumab • Hemodynamic and respiratory support

CRS, cytokine release syndrome; i.v., intravenous; TLS, tumor lysis syndrome.

18.2.3. ASTCT ICANS consensus grading scale for adults

ASTCT ICANS Consensus Grading for Adults

Neurotoxicity Domain	Grade 1	Grade 2	Grade 3	Grade 4
ICE score*	7-9	3-6	0-2	0 (patient is unarousable and unable to perform ICE)
Depressed level of consciousness†	Awakens spontaneously	Awakens to voice	Awakens only to tactile stimulus	Patient is unarousable or requires vigorous or repetitive tactile stimuli to arouse. Stupor or coma
Seizure	N/A	N/A	Any clinical seizure focal or generalized that resolves rapidly or nonconvulsive seizures on EEG that resolve with intervention	Life-threatening prolonged seizure (>5 min); or Repetitive clinical or electrical seizures without return to baseline in between
Motor findings‡	N/A	N/A	N/A	Deep focal motor weakness such as hemiparesis or paraparesis
Elevated ICP/ cerebral edema	N/A	N/A	Focal/local edema on neuroimaging§	Diffuse cerebral edema on neuroimaging; decerebrate or decorticate posturing; or cranial nerve VI palsy; or papilledema; or Cushing's triad

ICANS grade is determined by the most severe event (ICE score, level of consciousness, seizure, motor findings, raised ICP/cerebral edema) not attributable to any other cause; for example, a patient with an ICE score of 3 who has a generalized seizure is classified as grade 3 ICANS.
N/A indicates not applicable.

* A patient with an ICE score of 0 may be classified as grade 3 ICANS if awake with global aphasia, but a patient with an ICE score of 0 may be classified as grade 4 ICANS if unarousable.

† Depressed level of consciousness should be attributable to no other cause (eg, no sedating medication).

‡ Tremors and myoclonus associated with immune effector cell therapies may be graded according to CTCAE v5.0, but they do not influence ICANS grading.

§ Intracranial hemorrhage with or without associated edema is not considered a neurotoxicity feature and is excluded from ICANS grading. It may be graded according to CTCAE v5.0.

18.3. Encephalopathy assessment for children ≥ 12 years and adult

ICE
• **Orientation:** orientation to year, month, city, hospital: 4 points • **Naming:** ability to name 3 objects (eg, point to clock, pen, button): 3 points • **Following commands:** ability to follow simple commands (eg, "Show me 2 fingers" or "Close your eyes and stick out your tongue"): 1 point • **Writing:** ability to write a standard sentence (eg, "Our national bird is the bald eagle"): 1 point • **Attention:** ability to count backwards from 100 by 10: 1 point

Scoring: 10, no impairment;
7-9, grade 1 ICANS;
3-6, grade 2 ICANS;
0-2, grade 3 ICANS;
0 due to patient unarousable and unable to perform ICE assessment, grade 4 ICANS.

18.3.1. ASTCT ICANS consensus grading scale for children

ASTCT ICANS Consensus Grading for Children

Neurotoxicity Domain	Grade 1	Grade 2	Grade 3	Grade 4
ICE score for children age ≥12 years*	7-9	3-6	0-2	0 (patient is unarousable and unable to perform ICE)
CAPD score for children age <12 years	1-8	1-8	≥9	Unable to perform CAPD
Depressed level of consciousness†	Awakens spontaneously	Awakens to voice	Awakens only to tactile stimulus	Unarousable or requires vigorous or repetitive tactile stimuli to arouse; stupor or coma
Seizure (any age)	N/A	N/A	Any clinical seizure focal or generalized that resolves rapidly or nonconvulsive seizures on EEG that resolve with intervention	Life-threatening prolonged seizure (>5 min); or Repetitive clinical or electrical seizures without return to baseline in between
Motor weakness (any age)‡	N/A	N/A	N/A	Deep focal motor weakness, such as hemiparesis or paraparesis
Elevated ICP/ cerebral edema (any age)	N/A	N/A	Focal/local edema on neuroimaging§	Decerebrate or decorticate posturing, cranial nerve VI palsy, papilledema, Cushing's triad, or signs of diffuse cerebral edema on neuroimaging

ICANS grade is determined by the most severe event (ICE or CAPD score, level of consciousness, seizure, motor findings, raised ICP/cerebral edema) not attributable to any other cause. Baseline CAPD score should be considered before attributing to ICANS.

* A patient with an ICE score of 0 may be classified as grade 3 ICANS if awake with global aphasia, but a patient with an ICE score of 0 may be classified as grade 4 ICANS if unarousable.

† Depressed level of consciousness should be attributable to no other cause (eg, no sedating medication).

‡ Tremors and myoclonus associated with immune effector cell therapies may be graded according to CTCAE v5.0, but they do not influence ICANS grading.

§ Intracranial hemorrhage with or without associated edema is not considered a neurotoxicity feature and is excluded from ICANS grading. It may be graded according to CTCAE v5.0.

18.3.2. Encephalopathy assessment for children < 12 years

Encephalopathy Assessment for Children Age <12 Years Using the CAPD

Answer the following based on interactions with the child over the course of the shift					
	Never, 4	Rarely, 3	Sometimes, 2	Often, 1	Always, 0
1. Does the child make eye contact with the caregiver?					
2. Are the child's actions purposeful?					
3. Is the child aware of his/her surroundings?					
4. Does the child communicate needs and wants?					
	Never, 0	Rarely, 1	Sometimes, 2	Often, 3	Always, 4
5. Is the child restless?					
6. Is the child inconsolable?					
7. Is the child underactive; very little movement while awake?					
8. Does it take the child a long time to respond to interactions?					

(Adapted from Traube et al [36]; reproduced with permission.)
For patients age 1-2 years, the following serve as guidelines to the corresponding questions:
1. Holds gaze, prefers primary parent, looks at speaker.
2. Reaches and manipulates objects, tries to change position, if mobile may try to get up.
3. Prefers primary parent, upset when separated from preferred caregivers. Comforted by familiar objects (ie, blanket or stuffed animal).
4. Uses single words or signs.
5. No sustained calm state.
6. Not soothed by usual comforting actions, eg, singing, holding, talking, and reading.
7. Little if any play, efforts to sit up, pull up, and if mobile crawl or walk around.
8. Not following simple directions. If verbal, not engaging in simple dialog with words or jargon.

18.4. Management Algorithms for Immune Related Adverse Events

Early recognition and management of IRAE may mitigate severe toxicity. However, when an adverse event occurs you have to consider other potential etiologies than IRAE: disease relapse or progression or fortuitus event and eliminate other curable cause (*ie*: infection or thrombosis). Management algorithms have been proposed to assist investigators to detect and manage the following groups of IRAE:
Skin, gastrointestinal, pulmonary, hepatic, endocrinopathies, renal, neurological.

Toxicity (CTCAE v.5)	Management
Grade 1/2 Rash	• Symptomatic therapy (e.g. antihistamines, topical steroids) • Continue protocol therapy
Grade 3/4 Rash	Grade 3 : Hold infusion Grade 4 or grade 3 > 7 days : discontinue protocol therapy • Consider skin biopsy • Dermatology consult • 0.5-1 mg/kg/day methylprednisolone IV or IV equivalent

Toxicity (CTCAE v.5)	Management
Grade 1/2 AST or ALT increased	• Continue protocol therapy
Grade 3 or 4 AST or ALT increased	Hold infusion • Increase LFT monitoring to every 3 days If grade 3 > 7 days: Discontinue protocol therapy • Increase LFT monitoring to every 1-2 days • 1-2 mg/kg/day methylprednisolone IV or oral equivalent • Consult gastroenterologist

Toxicity (CTCAE v.5)	Management
Grade 1 bilirubin increased	• Continue protocol therapy
Grade 2 bilirubin increased	Hold infusion • Increase LFT monitoring to every 3 days
Grade 3 or 4 bilirubin increased	Grade 3: Hold infusion Grade 4 or grade 3 > 7 days: Discontinue protocol therapy • Increase LFT monitoring to every 1-2 days • 1-2 mg/kg/day methylprednisolone IV or oral equivalent • Consult gastroenterologist

Toxicity (CTCAE v.5)	Management
Grade 1 Diarrhea or Colitis	Continue protocol therapy • Symptomatic treatment

Grade 2 Diarrhea or Colitis	Hold infusion • symptomatic management. If persists for > 7 days, discontinue protocol therapy.
Grade 3/4 Diarrhea or Colitis	Grade 3 : Hold infusion Grade 4 or grade 3 > 7 days: Discontinue protocol therapy • 1 to 2 mg/kg/day methylprednisolone IV or IV equivalent • Consider prophylactic anti infectious treatment for opportunistic infections
	• Consider lower endoscopy

Toxicity (CTCAE v.5)	Management
Grade 1 Pneumonitis	Consider delay of protocol therapy • Monitor for symptoms ever 2-3 days • Consider Pulmonary and ID consults
Grade 2 Pneumonitis	Hold infusion • Pulmonary and ID consults • Monitor symptoms daily, consider hospitalization • 1 to 2 mg/kg/day methylprednisolone IV or oral equivalent • Consider bronchoscopy, lung biopsy
Grade 3 or 4 Pneumonitis	Discontinue protocol therapy • Hospitalize • Pulmonary and ID consults • 2-4 mg/kg/day methylprednisolone IV or IV equivalent • Consider prophylactic anti infectious treatment for opportunistic infections • Consider bronchoscopy, lung biopsy

Toxicity (CTCAE v.5)	Management
Grade 1 Neurological Toxicity	Continue protocol therapy
Grade 2 Neurological Toxicity	Hold infusion • Treat symptoms per local guidelines • Consider 0.5 to 1 mg/kg/day methylprednisolone IV or oral equivalent
Grade 3/4 Neurological Toxicity	Discontinue protocol therapy • Obtain neurology consult • Treat symptoms per local guidelines • 1-2 mg/kg/day methylprednisolone IV or IV equivalent • Consider prophylactic anti infectious treatment for opportunistic infections

Toxicity (CTCAE v.5)	Management
Asymptomatic TSH elevation	Continue protocol therapy • If TSH < 0.5 x LLN, or TSH > 2X ULN, or consistently out of range in 2 subsequent measurements: include fT4 at subsequent injection as clinically indicated • Consider endocrinology consult

Symptomatic Endocrinopathy	Evaluate endocrine function • Consider pituitary scan Symptomatic with abnormal lab/pituitary scan: • Hold infusion • 1-2 mg/kg/day methylprednisolone IV or oral equivalent • Initiate appropriate hormone therapy No abnormal lab/pituitary MRI scan but symptoms persist: • Repeat labs in 1-3 weeks/ MRI in 1 month
Suspicion of adrenal crisis (e.g. severe dehydration, hypotension, shock out of proportion to current illness)	Discontinue protocol therapy • Rule out sepsis • Stress dose of IV steroids with mineralocorticoid activity • IV fluids • Consult endocrinologist • If adrenal crisis ruled out, then treat as above for symptomatic endocrinopathy
Toxicity (CTCAE v.5)	Management
Grade 1 Creatinine Increased	Continue protocol therapy
Grade 2 or 3 Creatinine Increased	Hold infusion • Monitor creatinine every 2-3 days • 0.5-1 mg/kg/day methylprednisolone or oral equivalent • Consider renal biopsy
Grade 4 Creatinine Increased	Discontinue protocol therapy • Monitor creatinine weekly • 1 to 2 mg/kg/day methylprednisolone IV or IV equivalent • Consult nephrologist • Consider renal biopsy

18.5. Treatment of Nivolumab Related Infusion Reactions

- Since nivolumab contains only human immunoglobulin protein sequences, it is unlikely to be immunogenic and induce an infusion or hypersensitivity reaction. However, if such a reaction was to occur, it might manifest with fever, chills, rigors, headache, rash, pruritis, arthralgias, hypo- or hypertension, bronchospasm, or other symptoms.
- All Grade 3 or 4 infusion reactions should be reported within 24 hours to the Medical Monitor and reported as an SAE if criteria are met. Infusion reactions should be graded according to NCI-CTCAE 5.0 guidelines.
- Treatment recommendations are provided below and may be modified based on local treatment standards and guidelines as appropriate:

Grade 1

• Monitor patient until recovery from symptoms; infusion rate may be slowed.

• If the infusion is interrupted, then restart the infusion at 50% of the original infusion rate when symptoms resolve; if no further complications ensue after 30 minutes, the rate may be increased to 100% of the original infusion rate. Monitor patient closely.

The following prophylactic premedications are recommended for future infusions: diphenhydramine 1 mg/kg with max 50 mg (or equivalent, ie dexchlorpheniramine) and/or acetaminophen 10-15 mg/kg (max 1000mg) at least 30 minutes before additional nivolumab administrations, slowing infusion rate as above.

Grade 2

• Stop infusion, begin an IV infusion of normal saline, and treat the subject with diphenhydramine 1 mg/kg with max 50 mg IV (or equivalent, ie dexchlorpheniramine) and/or acetaminophen 10-15mg/kg (max 1000 mg); remain at bedside and monitor patient until resolution of symptoms. Corticosteroid or bronchodilator therapy may also be administered as appropriate.

• If the infusion is interrupted, then restart the infusion at 50% of the original infusion rate when symptoms resolve; if no further complications ensue after 30 minutes, the rate may be increased to 100% of the original infusion rate. Monitor patient closely.

• If symptoms recur, then no further nivolumab will be administered at that visit. Administer diphenhydramine 1 mg/kg with max 50 mg IV (or equivalent), and remain at bedside and monitor the patient until resolution of symptoms.

The following prophylactic premedications are recommended for future infusions: diphenhydramine 1 mg/kg with max 50mg IV (or equivalent, ie dexchlorpheniramine) and acetaminophen (10-15 mg/kg, max1000 mg) should be administered at least 30 minutes before additional nivolumab administrations. If clinically indicated, corticosteroids (recommended dose: 1-2 mg/kg/day methylprednisolone IV or equivalent) may be used.

Grade 3 or 4

Severe reaction, Grade 3: prolonged [ie, not rapidly responsive to symptomatic medication and/or brief interruption of infusion]; recurrence of symptoms following initial improvement; hospitalization indicated for other clinical sequelae [eg, renal impairment, pulmonary infiltrates]). Grade 4: (life- threatening; pressor or ventilatory support indicated).

• Immediately discontinue infusion of nivolumab

• Begin an IV infusion of normal saline, and treat the subject as follows. Recommend bronchodilators, epinephrine 0.2 to 1 mg of a 1:1,000 solution for subcutaneous administration or

0.1 to 0.25 mg of a 1:10,000 solution injected slowly for IV administration, and/or diphenhydramine 1 mg/kg with max 50 mg IV with 1-2 mg/kg/day methylprednisolone IV (or equivalent), as needed.

• Patient should be monitored until the investigator is comfortable that the symptoms will not recur.

• Nivolumab will be permanently discontinued.

Investigators should follow their institutional guidelines for the treatment of anaphylaxis. Remain at bedside and monitor patient until recovery from symptoms. In the case of late occurring hypersensitivity symptoms (*e.g.*, appearance of a localized or generalized pruritis within 1 week after treatment), symptomatic treatment may be given (*e.g.*, oral antihistamine or corticosteroids).

Please note that late occurring events including isolated fever and fatigue may represent the presentation of systemic inflammation. Please evaluate accordingly.